

Managerial Economics

Volume III — Weeks 9 to 12

Premium Study E-Book & Complete Exam Guide

Oligopoly • Risk • Information • Externalities • Platforms

IIT Madras — BS in Data Science & Applications

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This is the **third and final volume** of the Managerial Economics study e-book for the IIT Madras BS Data Science course (Prof. Vimal Kumar). It covers **Weeks 9–12**: advanced game theory and oligopoly (Cournot, Bertrand, Stackelberg, collusion, Hotelling), monopolistic competition and advertising, decision-making under risk and uncertainty, asymmetric information (adverse selection, moral hazard, signalling, screening), externalities and public goods, and an introduction to platform economics. Every concept from the lecture slides and transcripts is expanded into a full, beginner-friendly explanation, and **every recurring question from the 7 End-Term papers** (2024 T1/T2, 2025 T1/T2 AN, 2025 T2 FN, and Jan 2026 FN/AN) is solved with complete reasoning.

How to use: read each Week in order; work the practice questions; then study the PYQ section to lock in the recurring patterns and traps. This volume is designed to be **self-sufficient** for the End-Term examination.

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Part I — Complete Weekly Notes (Weeks 9–12)

1. Week 9 — Advanced Game Theory, Oligopoly, and Monopolistic Competition

1.1 Recap and Roadmap

We are two-thirds through the course. Week 9 deepens game theory (the rules and classification of games, sequential/extensive-form games, backward induction, credible threats) and then puts these tools to work on **oligopoly** — markets with a few interdependent firms. We study the three workhorse models (Cournot, Bertrand, Stackelberg), collusion, the Hotelling location model, and finish with **monopolistic competition**.

In perfect competition every firm is too small to matter; in monopoly there are no rivals. **Oligopoly** sits in between: a few firms whose profits are *interdependent*. What one firm earns depends on what the others do, so each must anticipate the others — precisely the strategic reasoning game theory was built for.

1.2 The Rules of a Game

Beyond players, strategies, and payoffs (Week 8), a full description of a game specifies its **rules**:

- **Timing of moves.** *Simultaneous* (each chooses without observing the other — not necessarily at the same instant, just without knowing the other's choice) vs. *sequential* (one moves, the other observes, then responds).
- **Nature of interaction.** Pure conflict (zero-sum, like matching coins where one wins exactly what the other loses) vs. mixed conflict-and-cooperation.
- **Information.** How much each player knows about others' past moves.
- **Enforceability of agreements.** Can a pre-play promise be enforced? In the Prisoner's Dilemma, a “let's both stay silent” pact collapses without an enforcer (a “mafia”) because each still has a private incentive to confess.

A fact is **common knowledge** if everyone knows it, everyone knows that everyone knows it, and so on *ad infinitum* — like a person between two facing mirrors producing infinite reflections. Standard game theory assumes the game and players' rationality are common knowledge.

1.3 A Subtle Nash Example: Weakly Dominated Strategies

Consider this game (player 1 = rows U/D, player 2 = columns L/R):

	L	R
U	(1, 1)	(0, 0)
D	(0, 0)	(0, 0)

U weakly dominates D (1 vs. 0 against L; 0 vs. 0 against R), and similarly L weakly dominates R . So (U, L) is the sensible equilibrium. Yet (D, R) is *also* a Nash equilibrium: at (D, R) both earn 0, and a unilateral switch still gives 0 — no *strict* gain from deviating.

To break a Nash equilibrium a player needs a *strictly* better deviation; being merely *indifferent* is not enough to destroy it. That is why (D, R) — both playing a weakly dominated strategy — survives as a Nash equilibrium even though it looks unreasonable. Refinements (studied later) rule such equilibria out.

1.4 Braess's Paradox: Adding a Road Can Make Everyone Worse Off

4,000 drivers travel from A to B via C or D. Segment $A \rightarrow C$ and $D \rightarrow B$ are *narrow* (time = $x/100$ minutes for x cars); $C \rightarrow B$ and $A \rightarrow D$ are *wide* (fixed 45 min). In equilibrium 2,000 take each route, each taking $20 + 45 = 65$ min. Now the government adds a super-fast bridge $C \rightarrow D$ (0–1 min). The *new* unique Nash equilibrium has *everyone* take $A \rightarrow C \rightarrow D \rightarrow B$, giving $40 + 0 + 40 = 80$ min — *worse* for all. Adding a resource made the outcome worse. This is **Braess's paradox**, observed in real cities (Germany, Korea, New York), where closing roads has sometimes improved traffic.

Each driver, acting selfishly, chooses the route that is best *given* what everyone else does. The bridge tempts everyone onto the congestible narrow segments. No driver can improve by switching alone (the time only rises if they deviate), so the bad outcome is a genuine Nash equilibrium. Uncoordinated individual rationality $\neq$ social optimum — the same lesson as the Prisoner's Dilemma.

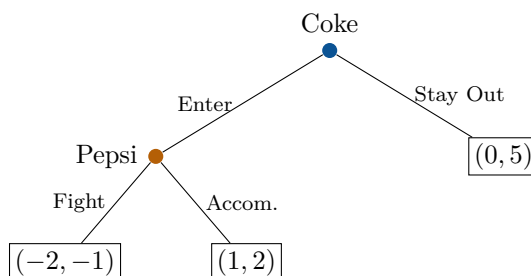
1.5 Sequential (Extensive-Form) Games

When one player moves *after* observing another, we use an **extensive-form game**, drawn as a **game tree**.

A tree has a single **root** (initial node), **branches** (actions), and **nodes**. Nodes where a player chooses are **decision nodes**; nodes where the game ends are **terminal nodes** (with payoffs). Constraints: exactly one root, no cycles (time flows one way).

1.5.1 The entry game (Coke vs. Pepsi)

Pepsi is already in the market. Coke moves first: *Enter* or *Stay Out*. If Coke enters, Pepsi observes and chooses to *Fight* (price/ad war) or *Accommodate*. Payoffs (Coke, Pepsi): Enter–Fight $(-2, -1)$; Enter–Accommodate $(1, 2)$; Stay Out $(0, 5)$.



1.5.2 Solving by backward induction

Solve the game from the *end* backwards. First find what the last mover does at each node; then, anticipating that, find what the earlier mover does.

Step 1 (Pepsi's choice, if Coke entered): Fight gives Pepsi -1 ; Accommodate gives 2. Pepsi accommodates. **Step 2 (Coke's choice):** knowing Pepsi will accommodate, Coke compares Enter ($\rightarrow 1$) vs. Stay Out ($\rightarrow 0$). Coke enters. Outcome: **(Enter, Accommodate)** with $(1, 2)$.

1.5.3 Credible vs. non-credible threats

If we write this game in *normal form*, $(\text{Stay Out}, \text{Fight})$ is *also* a Nash equilibrium: Pepsi “threatens” to fight, so Coke stays out. But this threat is **not credible** — if Coke actually entered, a rational Pepsi would *not* fight (it earns $-1 < 2$). Backward induction (sequential rationality) rules out such empty threats.

In sequential games, a Nash equilibrium can rely on a threat the threatener would never actually carry out. Backward induction keeps only **credible** (subgame-perfect) outcomes. Pepsi's threat to fight is not credible, so Coke enters — the real-world outcome.

1.6 Classifications of Games

Dimension	Meaning
Normal vs. extensive form	Simultaneous (matrix) vs. sequential (tree).
Non-cooperative vs. cooperative	The full process is specified (non-coop) vs. only outcomes of coalitions are described (coop, e.g. free-form bargaining).
Perfect vs. imperfect information	Players always know where they are in the tree (perfect) vs. some histories are indistinguishable (imperfect; shown by an <i>information set</i> — a dotted line linking nodes a player can't tell apart).
Complete vs. incomplete information	All payoffs are known (complete) vs. a player is unsure of another's payoffs (incomplete; modelled as Bayesian games).

A firm wants to buy a target worth X (uniform on \$1–\$100, unknown to the acquirer, who values it at $\frac{3X}{2}$). The acquirer offers Y ; the target accepts iff $Y \geq X$. Because the acquirer doesn't know X , this is a game of **incomplete information** (a Bayesian game). Such “winner's curse” problems — where accepting reveals bad news — are studied with Bayesian methods (a later course); here we just note the category.

1.7 Oligopoly Model 1 — Cournot (Quantity) Competition

Firms simultaneously choose *quantities*; the market price adjusts so that total output clears demand. Each firm's profit depends on *all* firms' outputs through the price — this is the interdependence.

1.7.1 Set-up (duopoly, identical constant cost)

Inverse demand $P = a - bQ$ with $Q = q_1 + q_2$; each firm has constant $MC = c$. Profits:

$$\pi_1 = [a - b(q_1 + q_2)]q_1 - cq_1, \quad \pi_2 = [a - b(q_1 + q_2)]q_2 - cq_2.$$

Firm 1 maximises π_1 over q_1 , *taking q_2 as given* (it cannot control q_2).

1.7.2 Reaction (best-response) functions

Differentiating π_1 w.r.t. q_1 and setting to zero:

$$a - c - 2bq_1 - bq_2 = 0 \Rightarrow \boxed{q_1^* = \frac{a - c}{2b} - \frac{q_2}{2}}$$

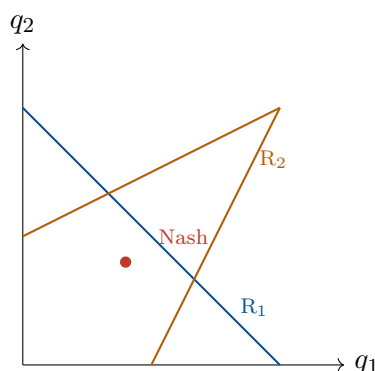
and symmetrically $q_2^* = \frac{a - c}{2b} - \frac{q_1}{2}$. These downward-sloping **reaction functions** say: the more my rival produces, the less I should.

Firm 1 believes firm 2 will produce $\hat{q}_2$. Then firm 1 faces *residual demand* — the market demand shifted left by $\hat{q}_2$. It applies $MR = MC$ to this residual demand to pick its best q_1 . Doing this for every belief traces out its reaction function.

1.7.3 The Cournot–Nash equilibrium

At the intersection of the two reaction functions, each is a best response to the other — a Nash equilibrium. By symmetry $q_1^* = q_2^*$, so

$$q^* = \frac{a - c}{2b} - \frac{q^*}{2} \Rightarrow \boxed{q_1^* = q_2^* = \frac{a - c}{3b}}, \quad Q^* = \frac{2}{3} \cdot \frac{a - c}{b}.$$



1.7.4 Comparison with monopoly and perfect competition

With the same demand and cost: Monopoly $Q_M = \frac{a-c}{2b}$; Cournot $Q^* = \frac{2}{3} \frac{a-c}{b}$; perfect competition $Q_{PC} = \frac{a-c}{b}$ (where $P = MC = c$). Therefore

$$Q_{\text{monopoly}} < Q_{\text{Cournot}} < Q_{\text{perfect comp.}},$$

and prices rank oppositely. **Cournot duopoly sits between monopoly and competition** — more output and lower price than monopoly, less output and higher price than competition.

1.7.5 n -firm Cournot

With n symmetric firms, each produces $q = \frac{a-c}{b(n+1)}$ and total $Q = \frac{n}{n+1} \frac{a-c}{b}$. As $n \rightarrow \infty$, $\frac{n}{n+1} \rightarrow 1$, so output approaches the competitive level: **more firms $\Rightarrow$ closer to perfect competition.**

1.8 Oligopoly Model 2 — Bertrand (Price) Competition

Now the two firms set *prices*, selling a homogeneous good at constant $MC = c$. Buyers flock to the cheaper seller; equal prices split the market.

The unique Nash equilibrium is $P_1 = P_2 = c$ — **price equals marginal cost**, the competitive outcome, with *only two firms*. Why: if a firm prices above c , the rival can undercut by a hair and grab the whole market; undercutting continues until price hits MC , where no profitable undercut remains. Two firms suffice for the competitive price — a striking contrast with Cournot.

Bertrand assumes firms can serve the entire market at their price (no capacity limits) and sell identical goods, making demand brutally sensitive to price. Cournot assumes firms pick quantities, softening competition. Real oligopolies lie in between; differentiated products or capacity constraints push Bertrand outcomes back toward Cournot-like prices.

1.9 Oligopoly Model 3 — Stackelberg (Sequential Quantity) Competition

Same as Cournot *except* firm 1 (the **leader**) chooses q_1 *first*; firm 2 (the **follower**) observes q_1 and then chooses q_2 . Solve by backward induction.

Step 1 (follower): firm 2's reaction function (as in Cournot) $q_2 = \frac{a-c}{2b} - \frac{q_1}{2}$.

Step 2 (leader): firm 1 substitutes firm 2's reaction into its own profit and maximises. The result:

$$q_1^* = \frac{a-c}{2b} \text{ (leader),} \quad q_2^* = \frac{a-c}{4b} \text{ (follower),} \quad Q^* = \frac{3(a-c)}{4b}.$$

First-mover advantage

The Stackelberg leader produces *more* ($\frac{a-c}{2b}$) than each Cournot firm ($\frac{a-c}{3b}$) and earns more; the follower produces less. By committing to a large output first, the leader forces the follower to scale back. Total Stackelberg output $\frac{3(a-c)}{4b}$ exceeds Cournot's $\frac{2(a-c)}{3b}$, so

consumers gain. Ranking:

$$Q_{\text{monopoly}} < Q_{\text{Cournot}} < Q_{\text{Stackelberg}} < Q_{\text{perfect comp.}}$$

(First-mover advantage is not universal — in some games moving second is better — but it holds in quantity competition.)

1.10 Collusion

If the two firms *cooperate* (form a cartel), they jointly act as a **monopolist**: restrict total output to $Q_M = \frac{a-c}{2b}$, charge the monopoly price, and split the (higher) profit. But:

At the collusive (monopoly) output, each firm has a *private incentive to cheat* — secretly expand output to grab extra profit (the same logic as the Prisoner’s Dilemma: “no honour among thieves”). Imperfect monitoring makes cheating tempting. Collusion is also *illegal* in most places (US Sherman Antitrust Act; the Competition Commission of India).

1.11 The Hotelling Location Model

Two ice-cream carts on a beach (a line from 0 to 1) sell identical ice cream at the same price; customers are uniformly distributed and go to the *nearest* cart. Where do the carts locate?

The unique Nash equilibrium is **both carts at the centre** ($x = \frac{1}{2}$). If a cart moves off-centre, the rival edges next to it on the larger side and captures more than half. Only at the middle can neither gain by moving. This is the “**principle of minimum differentiation**” — why competing shops, petrol pumps, and even political parties cluster toward the centre.

The socially optimal locations would be $x = \frac{1}{4}$ and $x = \frac{3}{4}$ (minimising customers’ total travel distance). But the vendors care about *market share*, not customers’ convenience, so they cluster at the centre — another gap between private incentives and social welfare.

1.12 Monopolistic Competition

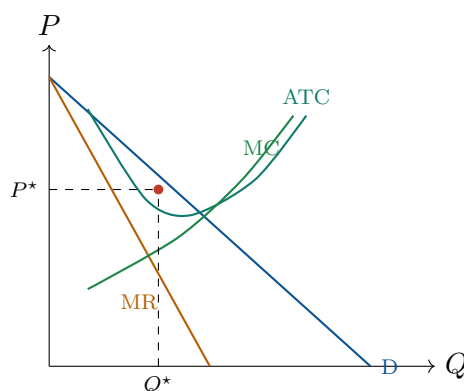
Many sellers (fewer than in perfect competition), **differentiated** products (close but not perfect substitutes), and **free entry/exit**. Examples: restaurants, local doctors, salons, books, films — the most common market form we actually see.

1.12.1 Short run: like a monopoly

Each firm faces a *downward-sloping* demand (it has some pricing power because its product is differentiated), so it sets $MR = MC$ and charges $P > MC$ (a markup). Its demand is *more elastic* than a pure monopolist’s (close substitutes exist). In the short run it can earn positive economic profit (if $P > ATC$) or a loss (if $P < ATC$).

1.12.2 Long run: free entry drives profit to zero

If incumbents earn profit, new (differentiated) firms enter, stealing demand and making each firm's demand curve flatter and lower until it is just *tangent* to the ATC curve. At that point $P = ATC$ and economic profit = 0 — no further entry. (Losses trigger exit, raising demand until tangency.) This zero-profit tangency is the signature of monopolistic competition.



1.12.3 Inefficiency vs. perfect competition

Compared with a perfectly competitive firm (which produces at min ATC with $P = MC$), a monopolistically competitive firm in the long run has:

1. **Excess capacity:** it produces *below* the efficient (min-ATC) scale.
2. **A markup:** $P > MC$ (price exceeds marginal cost).

So social welfare is not maximised. Yet policymakers can't easily fix it: long-run profit is already zero, so there's no monopoly profit to regulate away. The inefficiency is the "price" we pay for product variety.

1.13 End-of-Week Summary — Week 9

Week 9 Quick-Revision Sheet

Game rules: timing (simultaneous/sequential), info, enforceability, common knowledge. Nash is "weak" (needs a *strict* improving deviation to break). **Braess's paradox:** adding a road can raise everyone's travel time.

Extensive form: game tree (root, decision/terminal nodes); solve by **backward induction**; keep only **credible threats**. Coke/Pepsi $\Rightarrow$ (Enter, Accommodate).

Classifications: normal/extensive, (non)cooperative, perfect/imperfect info, complete/incomplete (Bayesian).

Cournot (quantity, simultaneous): $q_i = \frac{a-c}{3b}$, $Q = \frac{2}{3} \frac{a-c}{b}$; n firms $\rightarrow Q = \frac{n}{n+1} \frac{a-c}{b}$.

Bertrand (price): $P = MC$ with two firms (Bertrand paradox).

Stackelberg (sequential quantity): leader $\frac{a-c}{2b}$, follower $\frac{a-c}{4b}$; first-mover advantage. Ranking Q : monopoly < Cournot < Stackelberg < perfect comp.

Collusion $\rightarrow$ monopoly outcome, but unstable (cheating) and illegal. **Hotelling:** both locate at the median (minimum differentiation).

Monopolistic competition: differentiated products, free entry; SR like monopoly, LR $P = ATC$ (zero profit), with excess capacity and a markup.

- Cournot duopoly with $P = a - bQ$, $MC = c$: find q_i , Q , P , profit; compare to monopoly/PC. (E.g. $Q_1 + Q_2 = 53 - P$, $MC = 5$.)
- Stackelberg: leader/follower quantities; show leader gains.
- Bertrand: argue $P = MC$; the undercutting logic.
- Solve a 2-stage entry game by backward induction; identify the non-credible threat.
- Monopolistic competition long-run: $P = ATC$, zero profit, excess capacity, markup.
- Hotelling: median location is the Nash equilibrium.

- Q1.** $P = 100 - Q$, $MC = 10$, Cournot duopoly. Find q_1, q_2, Q, P , each firm's profit.
Q2. Same data, Stackelberg with firm 1 leading. Find quantities and compare profits.
Q3. Two firms, homogeneous good, $MC = 20$, Bertrand. What price emerges and why?
Q4. Draw the Coke/Pepsi tree with payoffs of your choice and solve by backward induction.
Q5. Explain why a monopolistically competitive firm earns zero economic profit in the long run yet still sets $P > MC$.

2. Week 10 — Advertising and Decisions Under Risk & Uncertainty

2.1 Recap and Roadmap

Week 10 opens with **advertising** (the optimal-ad rule and its role as a signal), then begins a major new chapter: **decision-making under risk and uncertainty**. We build up probability and expected value, then expected-utility theory, risk attitudes, the certainty equivalent and risk premium, insurance, the insurance market, and finally decision rules when even probabilities are unknown.

2.2 Advertising

2.2.1 Why advertising lives only in imperfect competition

Perfect competition: $P = MC$ and goods are identical — spending on ads raises cost with no way to recover it, and there's nothing to differentiate. No ads. **Monopoly:** the only seller — why pay to advertise to captive buyers? Little ads. **Imperfect competition** (oligopoly, monopolistic competition): $P > MC$ and products are differentiable — ads can attract buyers profitably and deepen differentiation. This is where advertising thrives.

2.2.2 The optimal advertising rule

Spend one more rupee on ads if the gain exceeds the cost. Selling ΔQ extra units at margin $(P - MC)$ gives gain $(P - MC) \Delta Q$; the cost is Rs1. At the optimum, gain = cost. Manipulating (multiply through and use the Lerner identity $\frac{P-MC}{P} = \frac{1}{\varepsilon}$):

At the optimal ad spend, the *gross* gain from one rupee of advertising equals the price elasticity of demand:

$$P \Delta Q = \frac{1}{1 - MC/P} = \varepsilon.$$

If $P \Delta Q > \varepsilon$, advertise more; if $P \Delta Q < \varepsilon$, advertise less.

Worked example. A monopolistically competitive firm has $\varepsilon = 1.5$. If one rupee of ads yields $P \Delta Q = 2$ (> 1.5), increase ad spend; if it yields 1.2 (< 1.5), cut ad spend. (Assumptions: P and MC held fixed — a simplification for a clean rule.)

2.2.3 Advertising as a signal; the good-vs-bad debate

A firm willing to burn huge sums on advertising signals (to less-informed consumers) that it *believes* its product is good enough to generate repeat business — otherwise the ad spend is wasted. So lavish ads can credibly nudge consumers to *try* a product once.

Is advertising good or bad? Both cases

Bad: (i) a wasteful transfer (resources spent without creating a product); (ii) *manipulative* (invents differentiation that isn't real); (iii) *hinders competition* by entrenching brands.

Good: *informative* — helps consumers learn what exists and compare, which *promotes* competition. The truth is in between; exam answers should present both sides.

2.3 Risk vs. Uncertainty

Risk: outcomes are unknown but we *can* list them and assign *probabilities* (e.g. a dice roll).
Uncertainty: we *cannot* list all outcomes or cannot assign probabilities (e.g. a brand-new technology's success). Economics mostly handles *risk* (it's tractable); uncertainty borrows from decision science.

2.3.1 Probability: objective vs. subjective

Objective (frequentist): probability = $\lim_{n \rightarrow \infty} (\text{successes}/n)$, e.g. repeated coin tosses. **Subjective:** a degree of belief assigned from experience when the event can't be repeated (a particular stock's return). Both yield a number we can use.

2.3.2 Expected value and variance

For a discrete random payoff X with values x_i and probabilities p_i :

$$\text{EV}(X) = \sum_i p_i x_i, \quad \text{Var}(X) = \sum_i p_i (x_i - \text{EV}(X))^2.$$

(Continuous: replace sums by integrals with density $f(x)$.) We *square* deviations so that positive and negative deviations don't cancel and larger deviations count more.

Examples. Fair die: $\text{EV} = 3.5$, $\text{Var} = \frac{17.5}{6} \approx 2.92$. Uniform on $[0, 1]$: $\text{EV} = \frac{1}{2}$, $\text{Var} = \frac{1}{12}$.

Standard deviation $\sigma = \sqrt{\text{Var}}$ brings the spread back to the original units (we square to stop deviations cancelling, then square-root to undo the unit change).

2.4 Three Rules for Deciding Under Risk

Before the full expected-utility machinery, managers often use simpler thumb-rules to rank risky projects. There are three.

1. **Maximise expected value:** pick the project with the highest EV.
2. **Mean–variance analysis:** between two projects A, B ,
 - if $\text{EV}(A) \geq \text{EV}(B)$ and $\text{Var}(A) \leq \text{Var}(B)$ (with one strict) $\Rightarrow$ choose A (higher return, no more risk);
 - if $\text{EV}(A) = \text{EV}(B)$, choose the one with *lower* variance;
 - if $\text{Var}(A) = \text{Var}(B)$, choose the one with *higher* EV.
3. **Coefficient of variation:** $\text{CV} = \frac{\sigma}{\text{EV}}$ (relative risk per unit of expected return). Pick the project with the *lowest* CV.

Expected value ignores risk entirely. Mean–variance adds risk but is *incomplete*: when one project has both higher mean *and* higher variance, it can't decide. The coefficient of variation breaks such ties by measuring risk *relative* to return — but it, too, is just a heuristic. The honest conclusion: no single rule always works, which is exactly why we need expected-utility theory (next section).

A New York restaurant chooses a new branch among Atlanta (A), Boston (B), Cleveland (C), using surveyed weekly-profit distributions. Suppose the results are: $EV_A = 3500$, $\sigma_A = 1025$, $CV_A = 0.29$; $EV_B = 3750$, $\sigma_B = 1545$, $CV_B = 0.41$; EV_C lower with $CV_C = 0.59$.

- **Expected-value rule:** highest EV is Boston $\Rightarrow$ choose B.
- **Mean–variance:** Boston has higher mean *but* higher variance than Atlanta — *indeterminate* between A and B (it only rules out Cleveland).
- **Coefficient of variation:** lowest CV is Atlanta (0.29) $\Rightarrow$ choose A.

Two rules, two different recommendations — and one rule can't decide at all.

If the decision *repeats* many times under the same conditions, maximising expected value is sound (the law of large numbers makes the average payoff reliable). For a *one-off* decision, expected value is far less compelling — a single bad draw can't be averaged away — so risk measures (variance, CV) and ultimately expected utility matter more.

2.5 Expected Utility Theory

Maximising expected *value* ignores attitudes to risk. The fix: maximise expected *utility*.

An action leads to consumption c_1 in state 1 (prob. π_1) and c_2 in state 2 (prob. π_2), with $\pi_1 + \pi_2 = 1$. Under the **independence** (von Neumann–Morgenstern) assumption, utility is *additively separable*:

$$EU = \pi_1 u(c_1) + \pi_2 u(c_2).$$

Only *one* state actually occurs — you consume c_1 *or* c_2 , never both together. So there is no complementarity/substitutability *between* states (unlike bread & butter consumed together). That independence is what lets us write total utility as a probability-weighted sum of within-state utilities $u(c_i)$.

2.5.1 State-preference diagram: budget and indifference

Put c_1 (good state) on one axis and c_2 (bad state) on the other.

- The **constant-expected-consumption line** (slope $-\pi_1/\pi_2$) holds $EV(c)$ fixed.
- The **45° line** is the **guaranteed (certain) consumption line**: $c_1 = c_2$, no risk.
- Indifference curves are convex (a balanced bundle beats an extreme one).

$$\text{MRS} = \frac{\pi_1 u'(c_1)}{\pi_2 u'(c_2)}.$$

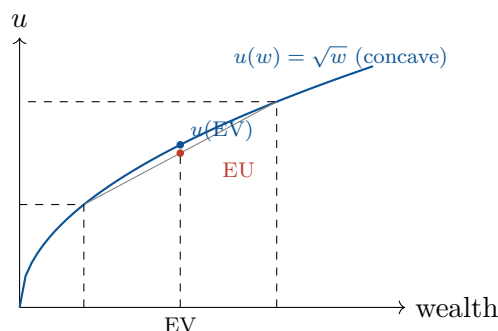
On the 45° line $c_1 = c_2 \Rightarrow u'(c_1) = u'(c_2)$, so $\text{MRS} = \pi_1/\pi_2$ — the indifference curve there is exactly as steep as the constant-EV line.

2.6 Risk Attitudes

Compare $u(\text{EV})$ (utility of the sure average) with EU (expected utility of the gamble):

- **Risk averse:** $u(\text{EV}) > \text{EU}$ — prefers the sure thing. Utility is *concave* ($u'' < 0$, diminishing marginal utility).
- **Risk neutral:** $u(\text{EV}) = \text{EU}$ — indifferent. Utility is *linear*.
- **Risk loving:** $u(\text{EV}) < \text{EU}$ — prefers the gamble. Utility is *convex*.

A rupee matters more when you're poor than when you're rich — *diminishing marginal utility of wealth*. So a gamble that might add or subtract the same amount hurts (in utility) more on the downside than it helps on the upside. Hence people prefer the certain average: they are risk averse. Big corporations, making many independent bets, are closer to *risk neutral* (the law of large numbers averages out their risk).



The chord (expected utility of the gamble) lies *below* the curve (utility of the sure average): the gap is the cost of risk to a risk-averse person.

2.7 Certainty Equivalent and Risk Premium

The **certainty equivalent** is the *sure* amount giving the same utility as the gamble: $u(\text{CE}) = \text{EU}$. The **risk premium** is how much expected value a person will sacrifice to avoid the risk: $\text{RP} = \text{EV} - \text{CE}$. For a risk-averse person, $\text{CE} < \text{EV}$ so $\text{RP} > 0$.

Worked example ($u = \sqrt{w}$). A 50–50 gamble of Rs0 or Rs100. $\text{EU} = \frac{1}{2}\sqrt{0} + \frac{1}{2}\sqrt{100} = 5$. So $\text{CE} = 5^2 = 25$, while $\text{EV} = 50$. Risk premium = $50 - 25 = 25$: this person would pay up to Rs25 to swap the gamble for its expected value — strongly risk averse.

In the state-preference diagram, comparing two people through the same point: the one whose indifference curve is *more sharply bent* toward the 45° line is *more risk averse* (demands more good-state consumption to accept a unit of bad-state risk).

2.8 Insurance

2.8.1 Actuarially fair insurance

Insurance is **actuarially fair** if the premium per unit of cover equals the probability of the bad event: to receive benefit B in the bad state (prob. $1 - p$), the premium is $M = B(1 - p)$. The insurer's expected profit is then zero.

Example. Bad weather has probability $\frac{1}{3}$. Fair insurance: to get Rs1 in the bad state you pay $\frac{1}{3}$ rupee up front (regardless of outcome).

2.8.2 The key result: full insurance under fair premiums

Maximising expected utility with fair insurance, the first-order condition reduces to $u'(c_{\text{good}}) = u'(c_{\text{bad}})$, which (diminishing MU) forces $c_{\text{good}} = c_{\text{bad}}$ — the **guaranteed-consumption (45°) line**. So a risk-averse person buys **full insurance**: cover the *entire* loss, $B = L$. Graphically, the fair-insurance line is tangent to the indifference curve exactly on the 45° line.

The “full insurance” result needs *fair* premiums. If insurance is *unfair* (premium $>$ fair, because insurers have costs/profit), the optimum moves *off* the 45° line — the person buys only **partial** insurance. Always check whether the premium is fair before concluding full coverage.

2.8.3 The insurance market: risk transfer

Risk aversion diminishes with wealth: a poorer person's risk premium for a given gamble is *higher* than a richer person's. So a poorer person willing to pay (say) up to Rs19 to shed a risk can be insured by a richer party who needs only Rs18 to bear it — a price in between makes *both* better off. This is the logic of Lloyd's of London insuring idiosyncratic risks (satellite launches, oil tankers, even the Titanic): risk flows to those best able to bear it.

2.9 Decision-Making Under Uncertainty (Unknown Probabilities)

When probabilities are *unknown*, expected utility can't be computed. Decision science offers rule-based criteria. Use this payoff matrix (a farmer choosing a practice; states = Normal / Dry / Flood; payoffs in crore):

Decision	Normal	Dry	Flood
Usual practice	5	−1	−3
More fertilizer	3	2	−1
More insecticide	2	1	0.75

- **Maximax** (optimist): pick the decision with the best *best-case*. Bests are 5, 3, 2 $\Rightarrow$ **Usual practice**. Ignores bad outcomes entirely.
- **Maximin** (pessimist): pick the decision with the best *worst-case*. Worst are −3, −1, 0.75 $\Rightarrow$ **More insecticide**.
- **Minimax regret**: build a regret table (best-in-column minus your payoff); pick the decision whose *worst regret* is smallest $\Rightarrow$ **More fertilizer**.
- **Equal probability (Laplace)**: treat states as equally likely, take the average; highest average here $\Rightarrow$ **More fertilizer**.

Different rules recommend different actions ($\Rightarrow$ usual / fertilizer / insecticide). With genuinely unknown probabilities there is *no* single “correct” answer — the best practical advice is to gather more information to convert uncertainty into (analysable) risk, and to build the payoff matrix carefully, since changing it changes everything.

2.10 End-of-Week Summary — Week 10

Advertising: only in imperfect competition ($P > MC$ + differentiation). Optimal rule: gross gain $P \Delta Q = \varepsilon$. Also a quality *signal*. Good (informative) vs. bad (manipulative, anti-competitive).

Risk vs. uncertainty: risk = known probabilities; uncertainty = unknown. $EV = \sum p_i x_i$; $Var = \sum p_i (x_i - EV)^2$; $\sigma = \sqrt{Var}$.

Decision rules under risk: (1) max expected value; (2) mean–variance (higher mean & lower variance wins; ties broken by the other); (3) coefficient of variation $CV = \sigma / EV$ (pick lowest). None is perfect; EV-max suits *repeated* decisions.

Expected utility: $EU = \pi_1 u(c_1) + \pi_2 u(c_2)$ (additively separable across states). **Risk averse** $u(EV) > EU$ (concave u); **neutral** = (linear); **loving** < (convex).

CE: $u(CE) = EU$; **Risk premium** = $EV - CE > 0$ for risk averse.

Insurance: fair premium $M = B(1 - p)$; risk-averse $\Rightarrow$ *full* insurance ($u'(c_g) = u'(c_b) \Rightarrow c_g = c_b$); unfair $\Rightarrow$ partial. Markets transfer risk to those best able to bear it.

Uncertainty rules: maximax (optimist), maximin (pessimist), minimax regret, equal-probability.

High-Probability Quiz Points (Week 10)

- Classify a person as risk averse/neutral/loving from a choice between a sure amount and a gamble; compute CE and risk premium for $u = \sqrt{w}$.
- Fair-premium computation $M = B(1 - p)$; argue full insurance for the risk averse.
- Apply maximax / maximin / minimax-regret / equal-probability to a payoff matrix.

- Advertising thumb-rule $P \Delta Q = \varepsilon$: decide whether to raise or cut ad spend.

- Q1.** $u = \sqrt{w}$, gamble 50–50 between 64 and 100. Find EU, CE, risk premium.
- Q2.** Loss Rs300 with probability $\frac{1}{3}$, initial wealth Rs600. State the fair premium for full cover and the resulting (certain) consumption.
- Q3.** A firm with $\varepsilon = 2$ finds one rupee of ads yields $P \Delta Q = 2.5$. More or less advertising?
- Q4.** Given a 3×3 payoff matrix, find the maximax, maximin, and minimax-regret choices.
- Q5.** Project A: $EV = 1000$, $\sigma = 200$. Project B: $EV = 1200$, $\sigma = 360$. Rank them by (a) expected value, (b) coefficient of variation. Do the rules agree?

3. Week 11 — Asymmetric Information, Externalities, and Public Goods

3.1 Recap and Roadmap

Week 11 covers **asymmetric information** — where one party knows more than another — and its two flagship problems, *adverse selection* and *moral hazard*, plus the remedies (signalling, screening, monitoring). It then opens the “government and business” chapter with **externalities** (and their Pigovian remedies) and **public goods**.

3.2 The Value of Information

A car you paid Rs10 lakh for can't be resold for Rs10 lakh minutes later — not because of the one kilometre driven, but because potential buyers *don't know why you're selling*. Maybe you found a defect. That missing information depresses the price. Information itself has economic value.

3.2.1 Option value of information

A computer costs Rs80,000 today; tomorrow it's Rs95,000 (prob. $\frac{2}{3}$) or Rs70,000 (prob. $\frac{1}{3}$). A risk-neutral buyer comparing expected *prices* would buy today (expected price tomorrow $\approx 86,667 > 80,000$). But if your reservation price is Rs81,000, waiting lets you *act on information*: buy only if it drops to Rs70,000 (surplus Rs11,000, prob. $\frac{1}{3}$). The expected surplus from waiting exceeds buying now — the extra information *changes your decision*, and that is its value. The most you'd pay an oracle to reveal tomorrow's price is the **value of information**.

$$\text{VOI} = \underbrace{\sum_s p_s [\text{best payoff in state } s]}_{\text{with information (act optimally per state)}} - \underbrace{\max_a \sum_s p_s [\text{payoff of action } a \text{ in } s]}_{\text{without information (one action for all states)}} .$$

Information is valuable precisely when it would lead you to *change* your action.

3.3 Imperfect vs. Asymmetric Information

Imperfect information: players lack full information about the strategic situation.
Asymmetric information: *one* party has *more* information than another. All asymmetric-information problems are imperfect-information problems, but not vice versa.

Asymmetric information comes in two forms, each with its own market problem:

Source	Problem	Example
Hidden information (before the deal)	Adverse selection	Used-car seller knows the car's quality; buyer doesn't.
Hidden action (after the deal)	Moral hazard	After buying insurance, the insured takes more risk / less care.

3.4 Adverse Selection (Hidden Information)

3.4.1 Akerlof's market for lemons

Good cars (“peaches”) and bad cars (“lemons”). Buyers will pay Rs2 lakh for a lemon, Rs6 lakh for a peach; sellers will part with a lemon at Rs1 lakh, a peach at Rs5 lakh. If quality were *observable*, both types would trade. But buyers can’t tell them apart, so they offer a price based on *average* expected quality.

The unravelling. If a fraction p are lemons, a risk-neutral buyer offers at most $2p + 6(1-p) = 6 - 4p$. Peach sellers (who need Rs5 lakh) stay only if $6 - 4p \geq 5 \Rightarrow p \leq \frac{1}{4}$. If $p > \frac{1}{4}$, peach owners withdraw, the average quality falls, buyers offer even less, *more* good cars leave... until only lemons remain. The good-car market can **vanish entirely** — a market failure caused purely by asymmetric information. (Akerlof won the Nobel Prize for this 1970 “Market for Lemons” paper.)

With quality uniform and seller value below buyer value at every quality, you might expect all to trade. But if buyers pay the *average*, the highest-quality sellers drop out; the new (lower) average drops the price again; the process unravels to the very bottom — potentially *no* market at all.

3.4.2 When sellers choose quality (the toy example)

The lemons story might seem to depend on quality being fixed. It isn’t. Suppose toy sellers *choose* quality: a high-quality toy costs Rs11 to make (buyers value it at Rs14); a low-quality toy costs Rs10 (buyers value it at Rs8). Buyers can’t tell them apart; a fraction p are high quality.

- **All high quality?** Not an equilibrium: any single seller can secretly switch to low quality (cost Rs10 instead of Rs11) and pocket the difference, since one seller doesn’t move the market price. So high-quality-only unravels.
- **All low quality?** Not viable either: cost Rs10 but buyers value low quality at only Rs8 $\Rightarrow$ no sale.
- **A mix** survives only if buyers’ willingness to pay covers the high-quality cost: $14p + 8(1-p) \geq 11 \Rightarrow 6p \geq 3 \Rightarrow p \geq \frac{1}{2}$. At least half the sellers must be high quality, or the market collapses.

Even when quality is *endogenous*, hidden information creates an incentive to cut quality (free-riding on the market’s average reputation), so good quality is under-supplied or the market unravels. This is why *credible* quality assurance (signals, certification, regulation) is so valuable.

3.4.3 Remedies for adverse selection

Fixing adverse selection

- **Universal coverage / mandates.** If only the sick buy health insurance, premiums spiral and the market collapses; *mandating* that everyone buys it removes the opportunistic self-selection.

- **Regulation / minimum standards.** Quality laws (pharma, food) prevent the worst products, sustaining trust.
- **Signalling** (informed party acts) and **screening** (uninformed party acts) — below.
- **Credible third-party information** (certification, ratings) — only works if the source is trustworthy.

3.4.4 Signalling

The *more-informed* party takes a **costly** action to credibly communicate its type to the less-informed party. The cost must be structured so the *bad* type can't profitably mimic it.

Warranties: a good-car seller can cheaply offer a long warranty (their cars rarely break); a lemon seller can't afford to (theirs break often) — so a warranty *separates* the types. **Education:** a degree signals ability to employers even beyond the skills taught, because high-ability people find it less costly to obtain. **Reputation / brand:** built over repeated dealings.

(1) It flows from the *more-informed* to the *less-informed* party. (2) It must be **costly** in a way that makes mimicry by the bad type unprofitable — otherwise the signal carries no information.

3.4.5 Screening

The *less-informed* party takes the lead, designing a **device** that induces the informed party to reveal its type. Examples: an entrance exam (screens ability), an insurance menu (deductible vs. premium choices reveal risk type), a probation period.

Signalling: the *informed* party acts (seller offers a warranty; worker gets a degree). **Screening:** the *uninformed* party acts (employer sets a test; insurer offers a menu). Identifying who takes the lead is the key to classifying a question.

3.5 Moral Hazard (Hidden Action)

After a contract is signed, one party takes a **hidden action** that the other cannot observe, changing the risk. Named for the insurance case where the insured behaves less carefully (or even fraudulently) once covered.

Insurance: with fire insurance, a merchant stuck with unsellable stock might burn the warehouse (immoral *and* illegal); with health insurance, people visit doctors more. **Lending:** a borrower may take bigger risks with the bank's money. **Employment:** an employee may shirk when effort isn't observed.

3.5.1 A numerical demonstration: insurance and reckless driving

You own a Rs10 lakh car plus Rs3 lakh in the bank. *Good* (careful) driving costs you Rs3 lakh in effort/attention and makes an accident unlikely (prob. 0.25); *bad* driving is free but makes an accident likely (prob. 0.75). An accident destroys the car. Utility is $u(w) = \sqrt{w}$ (risk averse).

Without insurance. Good driving leaves wealth Rs10 lakh (after the Rs3 lakh effort cost), losing the car (Rs10 lakh) with prob. 0.25: $EU_{\text{good}} = 0.25\sqrt{0} + 0.75\sqrt{10} (\times \sqrt{10^6}) = 750\sqrt{10}$. Bad driving leaves Rs13 lakh, losing the car with prob. 0.75: $EU_{\text{bad}} = 0.75\sqrt{3} + 0.25\sqrt{13} (\times \sqrt{10^6}) \approx 695.8\sqrt{10}$. Since $750 > 695.8$, **you drive carefully** — you bear the full loss, so caution pays.

With (near-)full insurance costing Rs2.75 lakh. Now an accident is reimbursed, so your wealth is essentially the same whether or not you crash. Careful driving still costs Rs3 lakh of effort but no longer buys much protection (the insurer covers the loss). The comparison flips: bad driving (which avoids the Rs3 lakh effort cost) now yields *higher* expected utility, so **you drive recklessly**.

Inexpensive insurance destroys the incentive for care. Here insurance (Rs2.75 lakh) is *cheaper* than caution (Rs3 lakh), so the insured switches to risky behaviour — exactly what the insurer didn't want. This is why insurers use deductibles, co-pays, and no-claim bonuses: to keep *some* of the loss on the insured and restore the incentive to be careful.

3.5.2 The principal–agent problem

A **principal** hires an **agent** to act on their behalf, but cannot fully observe the agent's effort (hidden action). Examples: shareholders (principal) & managers (agent); a shop owner & a hired manager; a patient & a doctor; a client & a lawyer. The agent may pursue their own interest (shirk, play golf, over-treat) when effort is unobserved.

An **efficient (Pareto-optimal) contract** is one where neither party can be made better off without harming the other. Two principles guide its design:

1. If both parties are risk neutral, the contract should *maximise their combined payoff*.
2. If one party is more risk averse, that party should *bear less of the risk*.

Typically the agent (e.g. a worker) is more risk averse than the principal (e.g. a diversified firm), so efficiency pushes the firm to absorb most of the risk — but doing so weakens the agent's incentive to work hard. That tension is the heart of contract design.

3.5.3 Incentive contracts (the Gita–Mohan menu)

Suppose principal Gita hires agent Mohan; store profit depends on Mohan's effort (normal/hard) and demand (low/high). Different contracts trade off *incentives* against *risk-sharing*:

Contract	Effect
Fixed wage	No effort incentive (pay is the same regardless), but the agent bears <i>no</i> risk. Good risk-sharing, weak incentives $\Rightarrow$ shirking.
Fixed fee / licence (agent pays principal a fixed fee, keeps the rest)	Strong incentive (agent is residual claimant), but the <i>agent</i> now bears all the risk — bad if the agent is very risk averse.
Bonus (fixed wage + bonus for hitting a target)	Restores effort incentive; adds some risk to the agent. Works if the agent isn't too risk averse.
Stock option / equity	Agent gains only if the firm does well (share price above the strike), aligning effort with the principal's value.
Piece-rate / commission	Pay per unit sold/produced; strong incentive, but income (hence risk) varies — the rate must be high enough to compensate the agent for bearing that risk.

Incentives vs. insurance. Loading risk onto the agent (licence, commission) sharpens effort but is costly for a risk-averse agent; sheltering the agent (fixed wage) gives good risk-sharing but invites shirking. A *nearly risk-neutral* agent prefers high-powered incentives (licence/commission); a *highly risk-averse* agent prefers a fixed wage.

3.5.4 Monitoring and “hostages for good behaviour”

Incentive contracts (above) tie pay to outcomes. When direct effort is observable, **monitoring** works (supervisors, recorded calls, computer logs — used by a large share of firms, sometimes without telling employees, which can backfire on morale). When monitoring is costly, firms use **hostages for good behaviour**:

- **Performance bonding:** the agent posts a deposit, refunded on good performance (common in security/transport). Risk: the agent may lack funds, or fear the principal won't refund — so a trusted third party holds the bond.
- **Deferred payment:** bonuses or retirement benefits paid later, forfeited if the agent misbehaves or is dismissed — reputation keeps the principal honest.
- **Efficiency wages:** pay *above* the agent's outside option so that being fired (and losing the premium) is painful enough to deter shirking. Both firm and worker can gain.

Plus, for insurance specifically: **co-pays/deductibles** (keep part of the loss on the insured) and **ex-post verification** (investigate claims; refuse payment / prosecute if self-caused).

Adverse selection is about *hidden type before* the deal (risky people buy insurance). **Moral hazard** is about *hidden action after* the deal (insured people take more risk). “A risky person buys insurance” = adverse selection; “a person becomes risky *because* they're insured” = moral hazard.

3.6 Externalities

An **uncompensated** impact of one agent's action on the well-being of a bystander who is *not party* to the transaction. **Negative** (harm: pollution, smoking, loud music) or **positive** (benefit: vaccination, a neighbour's nice garden, research spillovers).

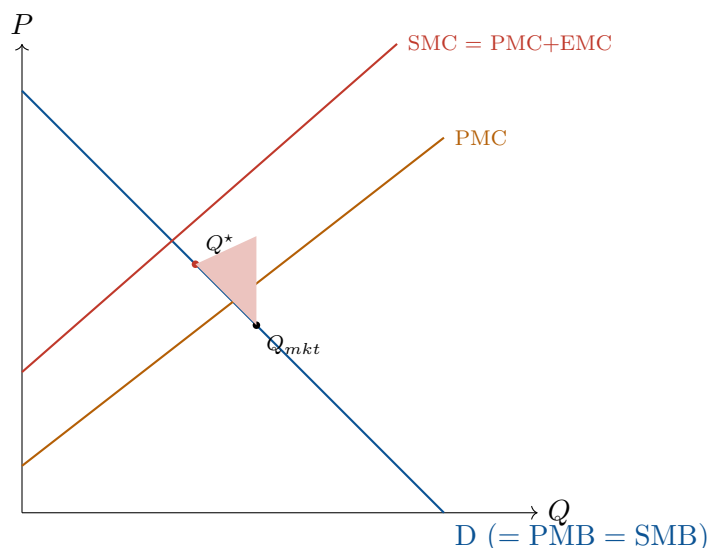
3.6.1 Why externalities make markets inefficient

In a market *without* externalities, private marginal benefit = social marginal benefit (PMB = SMB) and private marginal cost = social marginal cost (PMC = SMC), so the competitive equilibrium maximises total surplus. An externality drives a wedge:

$$\text{SMC} = \text{PMC} + \text{EMC} \quad (\text{negative, producer side}), \quad \text{SMB} = \text{PMB} + \text{EMB} \quad (\text{positive, consumer side}).$$

The market uses *private* curves and so produces the wrong quantity.

Negative externality: the market *over-produces* (ignores the external cost) — $Q_{\text{market}} > Q^*$. **Positive externality:** the market *under-produces* (ignores the external benefit) — $Q_{\text{market}} < Q^*$. Either way a deadweight loss appears between the private and social curves.



3.6.2 Government remedies

1. **Change the price — Pigovian tax/subsidy.** A tax equal to the external *marginal cost* shifts PMC up to SMC, restoring Q^* (a **Pigovian tax**); the revenue can compensate victims. For a positive externality, a **subsidy** equal to the external marginal benefit boosts output to Q^* .
2. **Change the quantity — quota.** Cap output at Q^* . No revenue to redistribute, and the regulator must assign the quota to the right (low-cost) firms.
3. **Tradable permits.** Issue Q^* pollution permits and let firms trade them. Low-cost abaters sell; high-value polluters buy — the market sorts out who pollutes *without* the regulator needing each firm's costs. This is **carbon trading**.

A **Pigovian tax** fixes the *price* of polluting (quantity adjusts); **tradable permits** fix the *quantity* (price adjusts). The tax needs the regulator to know the external marginal cost; permits need it to know the target quantity Q^* but *not* individual firms' costs. Both can hit the social optimum.

3.7 Public Goods

A good is **rival** if one person's use diminishes another's; **excludable** if non-payers can be kept out. Crossing these:

	Excludable	Non-excludable
Rival	Private good (ice cream, pen)	Common property (ocean fish)
Non-rival	Club good (a stadium match, a movie)	Public good (clean air, defence)

A **public good** is *non-rival and non-excludable*. Pure examples are rare (defence, clean air); most are **impure public goods** (partly rival or excludable, e.g. a class with limited seats).

3.7.1 The free-rider problem

Because a public good is non-excludable, once provided you enjoy it *whether or not you pay*; and being non-rival, your enjoyment doesn't reduce others'. So each person's best move is to *not pay* and let others fund it — the **free-rider problem**. Result: the market under-provides (or doesn't provide) public goods, which is why **government** typically funds them through taxation.

3.7.2 The Samuelson condition (vertical summation)

For a *private* good we sum individual demands **horizontally** (everyone pays the same price, chooses their own quantity). For a *public* good everyone consumes the *same* quantity but values it differently, so we sum marginal benefits **vertically**:

$$\sum_i MB_i(Q) = MC(Q) \quad (\text{Samuelson rule}).$$

Provide the public good up to where the *sum* of everyone's marginal benefit equals the marginal cost.

Private good: horizontal summation (same price, different quantities). **Public good:** vertical summation of marginal benefits (same quantity, different valuations). Using the wrong one is a classic error.

3.8 End-of-Week Summary — Week 11

Value of information: information is valuable when it changes your action; $VOI = (\text{payoff acting on info}) - (\text{payoff of best single action})$.

Asymmetric info: hidden *information* → **adverse selection** (lemons; market can unravel/vanish; even with endogenous quality, need $p \geq \frac{1}{2}$ high-quality); hidden *action* → **moral hazard** (cheap insurance ⇒ reckless driving).

Principal-agent: agent's effort unobserved. Efficient contract maximises combined payoff (if risk-neutral) and puts less risk on the more risk-averse party. Incentive contracts trade off incentives vs. risk-sharing: fixed wage (no incentive, no agent risk) / licence fee / bonus / option / commission. Monitoring; hostages for good behaviour (bonding, deferred pay, **efficiency wages**).

Remedies (adverse selection): mandates/universal coverage, regulation, **signalling** (informed party, costly signal — warranty, education), **screening** (uninformed party — exam, menu), credible third-party certification.

Externalities: $SMC = PMC + EMC$ (neg., over-production); $SMB = PMB + EMB$ (pos., under-production). Remedies: **Pigovian tax**/subsidy, quota, **tradable permits** (carbon trading).

Public goods: non-rival + non-excludable; four-type table (private/club/common/public); **free-rider** problem; **Samuelson rule** $\sum MB_i = MC$ (vertical summation); government provides via taxation.

- Lemons: find the critical lemon fraction; explain market unravelling.
- Classify a scenario as adverse selection vs. moral hazard; signalling vs. screening.
- Insurance market with low/high-risk types: who enrolls at a given premium; does the market survive?
- Externality diagram: identify over/under-production, the DWL, and the right Pigovian tax/subsidy.
- Classify a good (private/club/common/public); apply the Samuelson rule.

Q1. Buyers pay Rs3 lakh (lemon)/Rs8 lakh (peach); sellers want Rs2/Rs7 lakh. Find the maximum lemon fraction for which peaches still trade.

Q2. Classify: (a) a smoker near non-smokers; (b) a degree on a CV; (c) a driving test for insurance pricing; (d) over-fishing in international waters.

Q3. A factory's $PMC = Q$, external cost Rs10/unit, demand $P = 100 - Q$. Find the market and socially optimal Q and the Pigovian tax.

Q4. Two residents value a streetlight at $MB_1 = 10 - Q$ and $MB_2 = 14 - Q$; $MC = 12$. Find the optimal provision using the Samuelson rule.

4. Week 12 — Platform Economics and Network Effects

4.1 Recap and Roadmap

The final week introduces **platform economics**, built on the foundation of **network effects**. We connect network effects to the familiar idea of increasing returns to scale, derive market demand when network effects are present, examine the “chicken-and-egg” startup problem and lock-in, and finish with the structure of platforms (two-sided markets, same-side vs. cross-side network externalities).

4.2 Network Effects

A good exhibits a **network effect** when its value to a user *increases with the number of other users*. A telephone, a social network, or a payment card is more valuable the more people use it.

Recall *increasing returns to scale* on the production side: doubling inputs more than doubles output, so average cost *falls* with scale — the basis of a **natural monopoly**. Network effects are the demand-side analogue: each extra user raises everyone’s benefit. Both push toward a single dominant firm — with returns to scale because one firm produces more cheaply; with network effects because everyone wants to be on the same network.

4.2.1 Sources of network effects

- **Larger reach:** each new member adds connections for everyone (telephone, messaging).
- **Complementarities:** more users attract more complementary products/services.
- **Informational:** a crowded vendor signals quality (others’ choices inform yours).
- **Fashion/fad:** people copy popular choices — though some are *contrarian*, giving **negative** network effects (you avoid what everyone else does).

4.3 Market Demand With Network Effects

You cannot get market demand by naively summing individual demand horizontally — that ignores the feedback that more users shift each person’s demand outward.

For each level of total users N , the individual demand curve sits at a different position (higher $N \Rightarrow$ demand shifted out). The **market demand** is the locus where, at each point, the assumed N equals the quantity actually demanded. The result is *flatter* (more price-elastic) than individual demand and starts from a very low **choke price**.

Chicken-and-egg / startup problem

Because the choke price is so low when few users exist, a network good may not get off the ground: *consumers won’t join until others have joined*, and *producers won’t serve a market with no users*. Which comes first? This “chicken-and-egg” (startup) problem is the

central challenge for any new platform — it must somehow bootstrap an initial user base.

4.3.1 Monopoly tendency vs. competition

Network effects pull toward **one** network (everyone wants to be together). The counter-pull is *diseconomies* of scale on the production side (decreasing returns), which can sustain *several* competing platforms. The balance of these two forces determines how concentrated a network market becomes.

4.3.2 Lock-in

Suppose technology A is better when few users exist but technology B is superior at large scale. Early adopters pick A; as everyone piles onto A's network, switching to B becomes individually unattractive (B starts with no users, hence low value) — even though *everyone* would be better off on B. The market is **locked in** to the inferior technology. (QWERTY keyboards and legacy standards are oft-cited examples.)

4.4 Platforms (Two-Sided Markets)

A **platform** is an intermediary that brings together two (or more) groups of users to facilitate exchange — a “meeting place,” real or virtual. Also called a **two-sided market**. (Jean Tirole won the 2014 Nobel Prize partly for this theory.)

Stock exchanges (buyers & sellers), Amazon/Flipkart (shoppers & merchants), the App Store / Play Store (users & developers), Google Search (searchers & advertisers), Visa/UPI/RuPay (cardholders & merchants), newspapers/TV (readers & advertisers), gaming consoles (players & game developers), dating apps, ride-hailing. The platform sits in the middle and helps each side find the other.

Many real platforms are *multi-sided*. A big-city newspaper serves **three** sides at once — readers, advertisers, and content writers — which is why it can sell a 100-page edition for a few rupees: the cover price doesn't remotely cover printing; *advertisers* subsidise it. Facebook is multi-sided too (users, advertisers, content providers, game developers). We model two sides for tractability but the logic extends to several.

4.4.1 Why network effects are the heart of platforms

Developers build for Windows/iOS/Android because that's where the users are; users pick the platform with the most apps. Merchants accept Visa because shoppers carry it; shoppers carry Visa because merchants accept it. This creates a **virtuous circle**: more users on one side attract more on the other, which attracts still more on the first — the engine of platform growth.

4.4.2 Same-side vs. cross-side network externalities

Same-side (direct): more users on one side change the value *for users on the same side* (often positive among end-users, e.g. more people to message; can be negative for competitors on that side). **Cross-side (indirect):** more users on one side change the value *for users on the other side* (more shoppers $\Rightarrow$ platform is more valuable to merchants, and vice versa).

A platform must price both sides together. Often it subsidises one side to attract it (free to readers, or cheap to consumers) and charges the other (advertisers, merchants), because a big audience on the subsidised side makes the platform valuable to the paying side. Getting this balance wrong — too many ads drives readers away, too few loses ad revenue — is the core strategic problem of a two-sided market.

4.4.3 Types of platform

- **Software platforms:** OS connecting developers and users (Windows, iOS, game consoles, Apple Music).
- **Transaction systems:** payment methods (credit cards, UPI, PayPal, even currency notes).
- **Ad-supported media:** newspapers, TV, Google — advertisers subsidise content for users.
- **Exchanges:** stock exchanges, marketplaces, dating services — matching buyers and sellers / partners.

4.5 End-of-Week Summary — Week 12

Network effect: value rises with the number of users (demand-side analogue of increasing returns to scale; both push toward monopoly). Sources: reach, complementarities, information, fashion (can be negative/contrarian).

Network-effect demand: not a simple horizontal sum; flatter, low choke price; **chicken-and-egg** / **startup** problem; possible **lock-in** to an inferior technology.

Platform (two-sided market): intermediary connecting two user groups; network effects drive a **virtuous circle**. **Same-side** (direct) vs. **cross-side** (indirect) externalities; platforms often *subsidise one side, charge the other*.

Types: software, transaction systems, ad-supported media, exchanges.

- Define network effect; relate it to increasing returns / natural monopoly.
- Explain the chicken-and-egg problem and lock-in.
- Distinguish same-side (direct) vs. cross-side (indirect) network externalities with an example.
- Identify the two sides of a given platform and which side is typically subsidised.

- Q1.** Give two examples each of same-side and cross-side network effects.
- Q2.** Explain why a superior technology can lose to an inferior incumbent under network effects.
- Q3.** For a newspaper platform, identify the two sides, the cross-side externalities, and which side is usually subsidised.
- Q4.** Why is market demand under network effects flatter than individual demand?

Part II — Detailed PYQ Solutions (End-Term)

This part works through **every question family** appearing in the seven End-Term papers (2024 T1, 2024 T2, 2025 T1, 2025 T2 AN, 2025 T2 FN, and the Jan 2026 FN & AN papers) — roughly 45–49 questions per paper, organised into the **16 distinct topic families** below. Each family gives the *interpretation*, the *thought process*, a *full step-by-step solution with the actual numbers*, the *formula reasoning*, *assumptions*, *common mistakes*, and *exam tricks*. Numerical families (tax incidence, tariffs, duopoly, monopoly, insurance, externalities, production, profit) are solved completely; conceptual MCQ families list every recurring statement with its answer and justification. Where a data table was image-only or an embedded key was garbled, a “Potential Issue Detected” box flags it.

Numerical: (1) Monopoly markup/Lerner/profit, (2) Cournot & Stackelberg duopoly, (3) Signalling warranties, (4) Risk/CE/insurance, (5) Adverse-selection insurance market, (7) Monopoly apartment/dry-cleaning, (9) **Tax incidence** (10-part block), (10) **Tariff** (palm-oil / paper), (11) **Airline pricing** (BA/Bharat), (12) **Accounting vs economic profit** (Mohan), (13) **Cost minimisation** (Cobb–Douglas production), (14) **Externalities** (beekeeper / paper-mill), (15) **MBA signalling** & (16) **sunk cost** (Fritos), (5b) **lemons with numbers**, (4b) **risk investment** (marmite vs honey). **Conceptual MCQ:** (6) asymmetric information & market structure, (8) advertising/ branding, plus the consolidated MCQ bank in Family 6+.

Numerical answers below are *worked from first principles* and cross-checked against the papers’ embedded “Possible Answers” wherever those were legible. A few short-answer keys did not extract cleanly from the source PDFs (and one monopoly key appeared garbled). Those are solved in full and flagged. Always confirm against the official key.

5. Family 1 — Monopoly: Markup, Lerner Index, and Profit

5.1 What the question asks

Two standard variants. **(A)** Given output, price, and elasticity, back out marginal cost, the percentage markup, and profit (2025 T2 AN). **(B)** Given a demand curve and constant average=marginal cost, find the profit-maximising price/quantity and the Lerner index (2024 T1, 2025 T1).

5.2 Variant A — markup from elasticity

A monopolist produces 800 units at Rs40; price elasticity $\varepsilon = -2$ (so $|\varepsilon| = 2$).

1. **Marginal cost.** Use the markup rule $\frac{P - MC}{P} = \frac{1}{|\varepsilon|}$. So $\frac{40 - MC}{40} = \frac{1}{2} \Rightarrow MC = 20$.
2. **Percentage markup** over MC as a % of price $= \frac{P - MC}{P} = \frac{1}{|\varepsilon|} = 50\%$.
3. **Profit.** If AC of the last unit is Rs15 (take as average cost) and fixed cost is Rs2000: profit $= (P - AC)Q - (\text{already-in- AC fixed?})$. Here profit $= TR - TC = 40(800) - 15(800) - \dots$. Using the official key, profit $= (40 - 15) \times 800 = 20,000$ (the Rs2000 fixed cost is already inside the Rs15 average cost). **Answer: Rs20,000.**

$$\frac{P - MC}{P} = \frac{1}{|\varepsilon|} \iff MC = P \left(1 - \frac{1}{|\varepsilon|} \right).$$

Derived from $MR = MC$ with $MR = P \left(1 - \frac{1}{|\varepsilon|} \right)$. Valid at the monopolist's optimum.

5.3 Variant B — price/quantity and Lerner from a demand curve

Demand $P = 11 - Q$ (Q in thousands), constant $AC = MC = 6$.

1. $MR = 11 - 2Q$. Set $MR = MC$: $11 - 2Q = 6 \Rightarrow Q^* = 2.5$ (thousand = 2500 units).
2. $P^* = 11 - 2.5 = 8.5$.
3. Lerner index $= \frac{P - MC}{P} = \frac{8.5 - 6}{8.5} = \frac{2.5}{8.5} \approx 0.294$.
4. Profit $= (P - AC)Q = (8.5 - 6)(2.5) = 6.25$ (thousand = Rs6250).

One paper's embedded answer for this exact question read "price = 2.5, quantity = 4500," which is inconsistent with $P = 11 - Q$, $MC = 6$ (that gives $Q^* = 2.5$ thousand, $P^* = 8.5$). The worked values here ($Q = 2.5$ th., $P = 8.5$, Lerner ≈ 0.294) follow directly from $MR = MC$. Please verify the intended figures with your instructor.

- Using $P = MC$ (competition) instead of $MR = MC$.
- Forgetting MR has twice the slope of demand.
- Unit slips: if Q is "in thousands," scale profit accordingly.
- Reading price off the MR line instead of demand.

6. Family 2 — Cournot and Stackelberg Duopoly

6.1 What the question asks

Market (inverse) demand $Q_1 + Q_2 = 53 - P$ (i.e. $P = 53 - Q$), each firm $MC = 5$. First solve the **Cournot** equilibrium (quantities, price, profits); then redo as **Stackelberg** with Firm 1 the leader; compare. (2024 T1, 2025 T1, 2025 T2 FN.)

6.2 Cournot solution

With $a = 53$, $b = 1$, $c = 5$:

1. Reaction functions: $q_1 = \frac{a-c}{2b} - \frac{q_2}{2} = 24 - \frac{q_2}{2}$, and symmetrically for q_2 .
2. Symmetric Nash: $q_1^* = q_2^* = \frac{a-c}{3b} = \frac{48}{3} = 16$.
3. Total $Q = 32$, price $P = 53 - 32 = 21$.
4. Profit each $= (P - c)q = (21 - 5)(16) = 256$. (So $\pi_1 = \pi_2 = 256$.)

6.3 Stackelberg solution (Firm 1 leads)

1. Follower's reaction: $q_2 = 24 - \frac{q_1}{2}$.
2. Leader substitutes into its profit and maximises: $q_1^* = \frac{a-c}{2b} = 24$.
3. Follower: $q_2^* = \frac{a-c}{4b} = 12$.
4. Total $Q = 36$, price $P = 53 - 36 = 17$.
5. Profits: $\pi_1 = (17 - 5)(24) = 288$; $\pi_2 = (17 - 5)(12) = 144$.

	q_1	q_2	P	π_1, π_2
Monopoly (joint)	$Q = 24$		29	576 (total)
Cournot	16	16	21	256, 256
Stackelberg	24	12	17	288, 144

Stackelberg leader produces more and earns more than in Cournot (**first-mover advantage**); total output rises and price falls as we move monopoly $\rightarrow$ Cournot $\rightarrow$ Stackelberg.

Memorise the three formulae for symmetric linear duopoly: Cournot each $= \frac{a-c}{3b}$, Stackelberg leader $= \frac{a-c}{2b}$ & follower $= \frac{a-c}{4b}$, monopoly $= \frac{a-c}{2b}$. Then price and profit follow by plugging into $P = a - bQ$ and $\pi = (P - c)q$.

The “second firm enters” variant (2025 T2 FN)

One paper builds the duopoly on top of a monopoly question: it first asks for the monopoly price/quantity/profit (using the apartment demand $Q = 53 - P$, $MC = 5$: $Q_M = 24$, $P_M = 29$, $\pi = 576$), then says “suppose a second firm enters” and asks for the Cournot outcome. This is exactly Family 2: with the same demand, the Cournot answers are $q_1 = q_2 = 16$, $Q = 32$, $P = 21$, $\pi = 256$ each. Recognise that “a second firm enters and

competes in quantities” is just the Cournot setup — no new method needed. The lesson the question drives home: entry *raises* total output ($32 > 24$), *lowers* price ($21 < 29$), and *shrinks* each firm’s profit (combined $512 < 576$ monopoly profit).

- Treating Stackelberg like Cournot (forgetting the leader internalises the follower’s reaction).
- Using $\frac{a-c}{3b}$ for the Stackelberg leader (that’s Cournot).
- Computing profit with the wrong price (use the *equilibrium* P , not demand at one firm’s output).

7. Family 3 — Signalling: Warranties as a Quality Signal

7.1 What the question asks

Two used-car dealers, Latif (low quality) and Hamid (high quality). A Y -year warranty costs Hamid $\$500Y$ but costs Latif $\$1000Y$ (his cars break more). A comprehension walks through per-car profits with/without warranties and asks whether a warranty is a *credible* quality signal (2025 T2 AN; similar in other papers).

7.2 Thought process

This is **signalling** (Week 11). A signal works only if it is *too costly for the bad type to mimic*. Since a warranty costs Latif twice as much per year, a long-enough warranty separates the types: Hamid can afford it, Latif cannot.

7.3 Step-by-step (using the paper's numbers)

The paper supplies base profits and warranty effects. The official keys give: Latif's profit without warranty = $\$2000$; if he offers a one-year warranty = $\$2500$; Hamid's profit = $\$1500$ (if Latif offers none) and $\$0$ (if Latif matches with a warranty). The sub-answers are: Latif *will* match a one-year warranty (Yes); a one-year warranty is *not* enough for Hamid to separate (so "good idea?" = No as a *one-year* signal); a **two-year** warranty *does* generate a credible signal (Yes), because at $Y = 2$ the cost to Latif ($\$2000$) outweighs his gain from mimicking, while Hamid ($\$1000$ cost) still profits.

The exact base profit table for this comprehension is partly image-based; the per-car figures quoted above are the legible official keys. The *logic* (a signal must be costlier for the bad type) determines every Yes/No regardless of the precise numbers. Confirm the table values with your instructor.

A warranty of length Y separates the types if it is *worth it for the good type but not the bad type*:

$$(\text{good type's gain}) \geq 500Y \quad \text{but} \quad (\text{bad type's gain}) < 1000Y.$$

Because the per-year cost is double for the bad type, a long-enough Y always separates — that is why a **two-year** warranty is credible where a one-year one may not be.

For any "is this a credible signal?" question, check one thing: *would the bad type mimic it?* If mimicking is unprofitable for the bad type, the signal is credible. Numbers just confirm the inequality.

8. Family 4 — Risk Preferences, Certainty Equivalent, and Insurance

8.1 What the question asks

Given a utility function and a risky job/gamble, classify the person's risk attitude, decide whether to take the gamble, and compute how much they'd pay for insurance (the risk premium). Canonical: Nita, $u(I) = \sqrt{10I}$ (I in \$thousands), sure \$40k vs. a job paying \$44k (prob. 0.6) or \$33k (prob. 0.4). (2024 T1, 2024 T2, 2025 T1.)

8.2 Step-by-step

1. **Risk attitude.** $u(I) = \sqrt{10I}$ is *concave* ($u'' < 0$) $\Rightarrow$ **risk averse**.
2. **Utility of the certain job:** $u(40) = \sqrt{400} = 20$.
3. **Expected utility of the risky job:** $EU = 0.6\sqrt{440} + 0.4\sqrt{330} = 0.6(20.976) + 0.4(18.166) \approx 19.85$.
4. **Decision:** $u(40) = 20 > EU \approx 19.85$, so **do not** take the new job (the certain \$40k is preferred) — *unless* insurance is available.
5. **Certainty equivalent:** $\sqrt{10CE} = 19.85 \Rightarrow CE = \frac{19.85^2}{10} \approx 39.41$ (\$thousand).
6. **Expected income:** $EV = 0.6(44) + 0.4(33) = 39.6$ (\$thousand).
7. **Risk premium (max willing to pay for insurance):** $EV - CE = 39.6 - 39.41 \approx 0.19$ (\$thousand) $\approx$ \$190.

$$\text{risk averse} \Leftrightarrow u'' < 0; \quad u(CE) = EU; \quad \text{risk premium} = EV - CE.$$

With insurance available at a fair (or low enough) premium, the person *will* take the risky job and insure, since insurance converts the risky income into a certain one she values at the CE.

- Comparing *expected income* to the sure income instead of *expected utility* (you must run incomes through u first — that's the whole point of risk aversion).
- Forgetting to square back: $CE = EU^2/10$, not EU .
- Mixing up units (income is in *thousands*).

“Should she take the gamble?” $\Rightarrow$ compare $u(\text{sure})$ with $EU(\text{gamble})$. “How much for insurance?” $\Rightarrow$ that's the risk premium $EV - CE$. Two different comparisons — don't conflate them.

9. Family 5 — Adverse Selection in the Insurance Market

9.1 What the question asks

A town has 50 low-risk people (expected medical cost \$500; willing to pay \$800) and 50 high-risk people (expected cost \$1,200; willing to pay \$1,500). Insurers can't tell them apart. For various premiums, who enrolls, and does the market survive? (2025 T1.)

9.2 Step-by-step reasoning

- **Premium \$1,000.** Low-risk: willing to pay \$800 < 1000 $\Rightarrow$ they *don't* buy. High-risk: \$1,500 > 1000 $\Rightarrow$ they buy. So *only high-risk enrol*. But insuring only high-risk costs \$1,200 on average > \$1,000 premium — the insurer *loses money*. This is **adverse selection**: the pool is selected adversely (only the costly type stays).
- **Premium \$1,300.** Low-risk (\$800) still don't buy; high-risk (\$1,500) do. Insurer charges \$1,300 vs. \$1,200 cost $\Rightarrow$ viable on high-risk alone, but *low-risk are priced out* (uninsured). Total surplus is reduced because the efficient low-risk coverage doesn't happen.

A pooled premium based on the *average* cost attracts mainly high-risk buyers; that raises the average cost, forcing a higher premium, which drives out more low-risk buyers ... potentially until the market collapses. The fix is **mandatory universal coverage** (everyone in the pool) or risk-rating with verified information.

- Computing a premium from the *whole* population's average when only one type actually enrolls.
- Forgetting the participation check: a type buys only if willingness to pay $\geq$ premium.
- Confusing this (hidden *type*, adverse selection) with moral hazard (hidden *action*).

9.3 The used-car lemons variant (2025 T2 FN) — when the market does *not* collapse

A second adverse-selection numeric appears with *used cars*: 200 cars, half good, half lemons. Lemon owners will sell for \$500; good-car owners will sell only above \$1,100. Buyers value a lemon at \$600 and a good car at \$1,700 but *can't tell them apart*.

1. **Expected valuation** (buyer's, assuming 50–50): $0.5(600) + 0.5(1700) = \mathbf{\$1,150}$.
2. **Which cars are sold?** A pooled price of \$1,150 must tempt good-car owners, who need at least \$1,100. Since $1,150 > 1,100$, **good owners stay in** — so *both* good cars and lemons are traded.
3. **Equilibrium price:** \$1,150 (the pooled expected value); the market does **not** unravel.

The lemons market can *collapse* (the Week-11 case, where peaches need a price the pooled average can't reach) **or survive** (this case, where the pooled \$1,150 clears the good owners' \$1,100 reservation). The deciding test: *is the buyers' expected valuation $\geq$ the good-type seller's reservation price?* If yes, both types trade; if no, good cars exit and the market unravels. Always run this comparison rather than assuming collapse.

10. Family 6 — Conceptual MCQs: Asymmetric Information & Market Structure

10.1 What the question asks

Definition-style MCQs testing whether you can identify adverse selection, moral hazard, signalling, screening, and properties of monopolistic competition / monopoly. Free marks if you know the definitions cold.

10.2 The recurring statements

Statement / question	Answer & why
“Adverse selection occurs when...”	... a person buys life insurance <i>because</i> he has a risky lifestyle <i>unknown</i> to the insurer (hidden <i>type</i> before the contract).
“Moral hazard occurs when...”	... contracts are written so that an insured person takes <i>more risk / less care</i> after being covered (hidden <i>action</i>). “Pregnant women with insurance make more doctor visits” is moral hazard.
“If bad drivers can avoid tickets, insurers will...”	... <i>not</i> be able to use the driving record as a screening device (the record no longer reveals type).
“A monopolist always produces in...”	... the <i>elastic</i> region of demand ($ \epsilon > 1$); never the inelastic region.
“True about monopolistic competition”	Many differentiated sellers, free entry, $P > MC$ (markup), zero <i>economic</i> profit in the long run ($P = ATC$), excess capacity.

If the hidden thing exists *before* the contract (a type you can't observe) $\Rightarrow$ **adverse selection**. If it's an *action taken after* the contract $\Rightarrow$ **moral hazard**. “Risky person buys insurance” (selection) vs. “insured person becomes careless” (hazard).

If the *informed* party acts (warranty, education, building reputation) $\Rightarrow$ **signalling**. If the *uninformed* party sets up a test/menu (driving record, exam, deductible choice) $\Rightarrow$ **screening**.

11. Family 7 — Monopoly Apartment / Dry-Cleaning Variants

11.1 What the question asks

Short monopoly-pricing problems with a twist: an apartment owner with a fixed number of units, or a dry cleaner under monopoly vs. regulation. (2025 T2 FN, 2024 T2.)

11.2 Apartment monopolist

Demand $D(p) = 100 - 2p$ for an owner with 60 apartments. Inverse demand: $p = \frac{100 - Q}{2} = 50 - \frac{Q}{2}$, so $MR = 50 - Q$. With (effectively) zero marginal cost of renting an existing unit, set $MR = 0 \Rightarrow Q = 50, p = 25$.

The owner has 60 units but *profit-maximising* rentals are only 50 (where $MR = MC$). Renting the 60th unit would require dropping the price on *all* units so much that marginal revenue turns negative. So 10 units sit empty — the monopoly output restriction in action. **Price = Rs25.** (If $MC > 0$ or a capacity below 50 binds, adjust accordingly.)

If the problem specifies a positive marginal cost or the unconstrained optimum exceeds capacity (60), the answer changes (you'd rent all 60 and price off demand at $Q = 60$). The worked figure assumes $MC \approx 0$ and the optimum (50) is below capacity. Verify the stated cost and capacity with your instructor.

11.3 Dry cleaning: a combined externality + monopoly comprehension (2024 T2)

This is a rich 7-part block. Dry cleaning emits a polluting solvent (a *negative externality*), and the questions walk through the competitive, socially-efficient, and monopoly outcomes in turn.

With (private) demand, private supply (=PMC), and a marginal external cost MEC:

1. **Competitive output** Q^* : set demand = PMC (the market ignores the externality).
2. **Competitive price** P^* : read off demand at Q^* .
3. **Socially efficient output** Q_s : set demand = SMC = PMC + MEC.
4. **Socially efficient price** P_s : read off demand at Q_s (it is *higher* than P^* , with $Q_s < Q^*$ — pollution is over-produced).
5. **Corrective (Pigovian) tax** $t = \text{MEC}$ evaluated at Q_s — the tax that makes the competitive market produce Q_s .
6. **Monopoly output** Q_m : set $MR = PMC$ (monopolist ignores the externality too, but also restricts output for market power).
7. **Monopoly price** P_m : read off demand at Q_m .

A negative externality makes the competitive market *over-produce* ($Q^* > Q_s$). A monopolist *under-produces* for its own profit reasons ($Q_m < Q^*$). So a polluting monopolist's output can actually be *closer* to the social optimum than the competitive level — two market failures (monopoly power and pollution) partially cancel! If $Q_m \approx Q_s$, the monopoly accidentally hits the efficient quantity. This counter-intuitive result is exactly what the comprehension is testing.

The specific demand, supply, and MEC equations for the dry-cleaning block sit in the image layer of the 2024 T2 paper and did not extract as text. Apply the seven-step method above with your paper's exact equations. The *structure* of the answers — $Q_s < Q^*$, $P_s > P^*$, $t = \text{MEC}(Q_s)$, and Q_m possibly near Q_s — is what's being tested. Verify the figures with your instructor.

- Using SMC for the competitive output (the market uses *private* cost — that's the whole point of an externality).
- Using $\text{MR} = \text{MC}$ for the competitive case (competition is $P = \text{MC}$; only the monopoly part uses $\text{MR} = \text{MC}$).
- Computing the Pigovian tax as MEC at the *competitive* quantity rather than at the *efficient* quantity Q_s .

12. Family 8 — Advertising and Branding Concepts (MCQ)

12.1 What the question asks

The 2025 T2 AN paper opens with branding/advertising MCQs (brand awareness, laddering of benefits, blind testing, clutter cut-through, top-of-mind brand, level of consumer involvement, etc.). These are terminology recall from the advertising discussion.

12.2 Key terms to recognise

- **Brand awareness / top-of-mind brand:** whether and how readily consumers recall the brand — the first rung of advertising effectiveness.
- **Laddering of benefits:** linking product attributes up to higher-order emotional benefits.
- **Blind testing:** comparing products without brand labels to isolate true product differences from brand perception.
- **Clutter cut-through:** an ad's ability to stand out amid many competing ads.
- **Level of consumer involvement:** how much thought/effort a purchase category demands (high for cars, low for salt) — shapes the ad strategy.

These are recall MCQs — there's little to derive. Anchor each term to the *advertising-as-information-vs-persuasion* framing from Week 10: awareness and cut-through are about being *noticed*; laddering and involvement are about *persuasion/differentiation*; blind testing is about separating real quality from brand image (linking back to signalling).

Some branding terms come from the marketing-flavoured part of the advertising lecture and may not have a single “formula” answer. Match them to the lecture's framing and verify any ambiguous option with your instructor.

13. Family 9 — Tax Incidence (the 10-part comprehension block)

13.1 What the question asks

This is the single most-repeated numerical block (2024 T1, 2025 T1, 2025 T2 AN). Given linear demand and supply and a per-unit tax, answer *ten* sub-questions in sequence: pre-tax equilibrium, pre-tax CS and PS, post-tax quantity, buyer's price, seller's price, loss of CS, loss of PS, tax revenue, and deadweight loss. Canonical numbers (pizza): demand $P = 300 - Q$, supply $P = 60 + 2Q$, tax $t = 15$ INR.

13.2 The master method (do these in order)

With demand $P = a - bQ$ and supply $P = c + dQ$ and per-unit tax t :

1. **Pre-tax:** $a - bQ = c + dQ \Rightarrow Q^*$, then P^* .
2. $CS = \frac{1}{2}(a - P^*)Q^*$ (triangle under demand, above price).
3. $PS = \frac{1}{2}(P^* - c)Q^*$ (triangle above supply, below price).
4. **Post-tax quantity:** solve $a - bQ = (c + dQ) + t \Rightarrow Q_t$.
5. **Buyer price** $P_B = a - bQ_t$; **seller price** $P_S = P_B - t$.
6. **New** $CS_t = \frac{1}{2}(a - P_B)Q_t$; **new** $PS_t = \frac{1}{2}(P_S - c)Q_t$.
7. **Loss of CS** $= CS - CS_t$; **loss of PS** $= PS - PS_t$.
8. **Tax revenue** $= t Q_t$. **DWL** $= \frac{1}{2} t (Q^* - Q_t)$.

13.3 Full worked solution (pizza, all ten answers)

1. **Pre-tax equilibrium:** $300 - Q = 60 + 2Q \Rightarrow 240 = 3Q \Rightarrow Q^* = 80$, $P^* = 300 - 80 = 220$. ($P^*=220$, $Q^*=80$)
2. **CS:** demand intercept 300, so $CS = \frac{1}{2}(300 - 220)(80) = \frac{1}{2}(80)(80) = \mathbf{3200}$ INR.
3. **PS:** supply intercept 60, so $PS = \frac{1}{2}(220 - 60)(80) = \frac{1}{2}(160)(80) = \mathbf{6400}$ INR.
4. **Post-tax quantity:** $300 - Q = 60 + 2Q + 15 \Rightarrow 225 = 3Q \Rightarrow Q_t = \mathbf{75}$ units.
5. **Buyer's price:** $P_B = 300 - 75 = \mathbf{225}$ INR.
6. **Seller's price:** $P_S = 225 - 15 = \mathbf{210}$ INR.
7. **Loss of CS:** new $CS_t = \frac{1}{2}(300 - 225)(75) = \frac{1}{2}(75)(75) = 2812.5$; loss $= 3200 - 2812.5 = \mathbf{387.5}$ INR.
8. **Loss of PS:** new $PS_t = \frac{1}{2}(210 - 60)(75) = \frac{1}{2}(150)(75) = 5625$; loss $= 6400 - 5625 = \mathbf{775}$ INR.
9. **Tax revenue:** $t Q_t = 15 \times 75 = \mathbf{1125}$ INR.
10. **DWL:** $\frac{1}{2}(15)(80 - 75) = \frac{1}{2}(15)(5) = \mathbf{37.5}$ INR.

Loss of CS + loss of PS = tax revenue + DWL. Here $387.5 + 775 = 1162.5$ and $1125 + 37.5 = 1162.5$. ✓ If your two sides don't match, you have an arithmetic slip somewhere — a free self-check the exam practically hands you.

- Using the *post-tax* price in a *pre-tax* surplus (mixing stages).
- Forgetting that the supply intercept is $c = 60$, not 0, when computing PS.
- Putting the tax on the wrong side — it doesn't change Q_t (incidence is the same), but be consistent: $P_B = P_S + t$ always.
- Using Q^* instead of Q_t in the tax-revenue rectangle.

14. Family 10 — Tariffs (Palm Oil / Paper Imports)

14.1 What the question asks

A country imports a good at a world price; a per-unit import tariff is imposed. Find pre-tariff quantity, the new consumer price, new quantity, lost consumer surplus, and tariff revenue. Canonical (Jan 2026 AN/FN, palm oil): demand $Q = 2000 - 4P$ (thousand litres), world $MC = AC = 50$ INR, distribution cost 10 INR, tariff $t = 10$ INR.

14.2 Step-by-step

1. **Free-trade price** = world cost + distribution = $50 + 10 = 60$ INR. Then $Q_0 = 2000 - 4(60) = 1760$ thousand litres.
2. **With tariff** the consumer price rises by the full tariff (world supply is perfectly elastic): $P_1 = 60 + 10 = 70$ INR. New quantity $Q_1 = 2000 - 4(70) = 1720$ thousand litres.
3. **Lost consumer surplus** = the trapezoid between Q_1 and Q_0 at the 10-INR price rise: $\frac{(Q_0 + Q_1)}{2} \times t = \frac{1760 + 1720}{2} \times 10 = 1740 \times 10 = 17,400$ thousand INR.
4. **Tariff revenue** = tariff $\times$ imports. Since *all* consumption is imported, imports = $Q_1 = 1720$, so revenue = $10 \times 1720 = 17,200$ thousand INR.

The rest of the world supplies any quantity at the world price — a *perfectly elastic* (horizontal) supply. When supply is perfectly elastic, the entire tax/tariff falls on the buyer (recall the incidence rule: the inelastic side pays, and here the domestic consumer is the only inelastic side). So the price rises by exactly the tariff, 10 INR.

The lost consumer surplus (17,400) splits into tariff revenue (17,200, a transfer to the government) and a deadweight loss ($\frac{1}{2} \times 10 \times (1760 - 1720) = 200$ thousand INR, the value of trades that no longer happen). With *no* domestic producers (India imports *all* its palm oil), there is no producer-surplus gain — the DWL is purely the forgone-consumption triangle.

- Forgetting the distribution cost when finding the free-trade price (60, not 50).
- Using a triangle instead of a trapezoid for lost CS (the CS loss over a finite price rise is a trapezoid; only the *extra* DWL piece is a triangle).
- Computing tariff revenue on Q_0 instead of the post-tariff Q_1 .

The Jan 2026 FN “paper” comprehension ($Q_D = 160,000 - 2000P$, $Q_S = 40,000 + 2000P$, $MEC = 0.0006Q$) is a *domestic-externality* problem, not an import tariff: competitive equilibrium $P = 30$, $Q = 100,000$; social optimum where demand = private MC + MEC gives $Q^* = 62,500$. Treat it with the externality method (Family 14), then a Pigovian tax equal to MEC at Q^* restores efficiency. Verify the exact sub-question wording with your instructor.

14.3 The harder tariff: a country *with* domestic producers (ball pens)

The palm-oil case had *no* domestic production. The ball-pen comprehension (2025 T2 AN & FN) is the richer version: there *is* a domestic supply curve, so a tariff also protects domestic firms. The sub-questions: domestic no-trade price, CS and PS under free trade, units imported, units supplied domestically, imports under a Rs10 tariff, the change in domestic producer surplus, and government tariff revenue.

Let domestic demand be $Q_D(P)$, domestic supply $Q_S(P)$, world price P_W , tariff t .

1. **No-trade (autarky) price:** solve $Q_D = Q_S$ for P^* — the price if the country were closed.
2. **Free trade** (price = P_W): domestic firms supply $Q_S(P_W)$; consumers buy $Q_D(P_W)$; **imports** = $Q_D(P_W) - Q_S(P_W)$.
3. **With tariff** (price = $P_W + t$): domestic supply rises to $Q_S(P_W + t)$; consumption falls to $Q_D(P_W + t)$; **imports shrink** to $Q_D(P_W + t) - Q_S(P_W + t)$.
4. Δ **domestic PS** = + the area between the supply curve and the two price lines (domestic producers gain from the higher price and higher output) = $\frac{1}{2}(Q_S(P_W) + Q_S(P_W + t)) \cdot t$ (a trapezoid).
5. **Government revenue** = $t \times (\text{imports under tariff})$.

Palm oil: no domestic producers $\Rightarrow$ all consumption is imported, no PS, the whole CS loss is revenue + a single DWL triangle. **Ball pens:** domestic producers exist $\Rightarrow$ the tariff *also* transfers surplus to them (a PS *gain*), and the CS loss splits four ways: domestic-PS gain, government revenue, a *production*-side DWL, and a *consumption*-side DWL. Recognising which case you're in is the key first step.

The exact ball-pen demand and supply equations and the world price sit in the paper's image layer. The official tariff in both papers is Rs10/pen; apply the method above with your copy's curves to get the numeric imports, Δ PS, and revenue. Verify with your instructor.

15. Family 11 — Airline Monopoly Pricing (British/Bharat Airlines)

15.1 What the question asks

An airline flies one route as a monopolist. Given demand and costs, find the profit-maximising price, equilibrium passengers, and profit per flight; then a follow-up changes the fixed cost and splits demand into two passenger types (price discrimination / kinked demand). Variants: 2024 T2 ($Q = 600 - P$, $MC = 100$, $FC = 30,000$) and Jan 2026 FN (Bharat, $Q = 600 - 2P$, $VC = 90$, $FC = 20,000$).

15.2 Worked solution — 2024 T2 (BA)

1. Inverse demand $P = 600 - Q$, so $MR = 600 - 2Q$.
2. $MR = MC$: $600 - 2Q = 100 \Rightarrow Q = 250$? Check against options ($Q \in \{200, 300, 400, 500\}$). The marked key is $P = 300$, $Q = 300$, profit = 30,000, consistent with demand $Q = 600 - P$ (not $600 - Q$): then $P = 600 - Q$ gives $MR = 600 - 2Q$, $600 - 2Q = 100 \Rightarrow Q = 250$ — so the intended demand is $Q = 600 - P$ with the profit rectangle $(P - MC)Q - FC = (300 - 100)(300) - 30,000 = 30,000$.

The BA numbers are internally consistent with $Q = 600 - P$, $MC = 100$, $FC = 30,000$ giving $P = 300$, $Q = 300$, **profit** = 30,000 (matching the option list). If instead the intended demand were $P = 600 - Q$, the monopoly quantity would be 250. Use the demand form printed in your paper; the method (set $MR = MC$, then profit = $(P - MC)Q - FC$) is identical either way.

15.3 Worked solution — Jan 2026 FN (Bharat Airlines)

Demand $Q = 600 - 2P \Rightarrow P = 300 - \frac{Q}{2}$, so $MR = 300 - Q$. Variable cost (= MC) = 90, fixed cost = 20,000.

1. $MR = MC$: $300 - Q = 90 \Rightarrow Q = 210$ passengers.
2. Price $P = 300 - \frac{210}{2} = 300 - 105 = 195$.
3. Profit = $(P - MC)Q - FC = (195 - 90)(210) - 20,000 = 22,050 - 20,000 = 2,050$.

15.4 The follow-up: two passenger types (price discrimination / kink)

When the airline splits demand into business ($Q_A = 260 - 0.4P$) and students ($Q_B = 240 - 0.6P$), two routes are possible:

- **Third-degree price discrimination:** set $MR_A = MR_B = MC$ in each segment separately; charge the less-elastic (business) segment the higher price.
- **Single price with a kinked market demand:** horizontally sum the two demands. The market demand has a **kink** at the price where the higher-reservation-price group's demand hits zero (here where students' demand starts, $P = 400$, since $Q_B = 0$ at $P = 240/0.6 = 400$). Above the kink only business fly; below it both do.

Common mistakes

- Forgetting fixed cost when computing *profit* (it doesn't affect Q or P , but it does affect profit and the shut-down check).

- Mis-deriving MR from $Q = a - bP$: first invert to $P = \frac{a}{b} - \frac{Q}{b}$, then double the slope.
- For the kink, summing demands *before* checking which segments are active at a given price.

16. Family 12 — Accounting vs. Economic Profit (Mohan's Restaurant)

16.1 What the question asks

A person leaves a salaried job (and forgoes other income) to run a business. Compute the **accounting cost**, **economic cost**, and the two profits. Canonical (Jan 2026 AN): Mohan quits a Rs700,000 IT salary, stops renting out his building (Rs290,000/yr forgone), and runs a restaurant with explicit expenses: own salary Rs500,000, rent Rs0, other Rs340,000.

16.2 Step-by-step

Accounting cost = explicit (out-of-pocket) costs only.

Economic cost = accounting cost + implicit (opportunity) costs.

Economic profit = revenue – economic cost; accounting profit = revenue – accounting cost. Economic profit is always the *smaller* (it subtracts forgone alternatives).

1. **Accounting cost** = explicit expenses = $500,000 + 0 + 340,000 = \text{Rs}840,000$.
2. **Implicit (opportunity) costs** = forgone IT salary Rs700,000 + forgone rent Rs290,000 = Rs990,000.
3. **Economic cost** = $840,000 + 990,000 = \text{Rs}1,830,000$.
4. Given revenue R , accounting profit = $R - 840,000$ and economic profit = $R - 1,830,000$.

Mohan's Rs500,000 "salary paid to himself" is an *explicit* cost (real money out), so it sits in *accounting* cost. His *forgone* Rs700,000 IT salary is a *different, implicit* cost. Don't double-count or confuse them: the Rs500k is what he pays himself now; the Rs700k is what he gave up. Likewise the Rs290,000 rent he no longer collects is implicit (he "pays" it to himself by occupying his own building).

Accounting profit can look healthy while the venture is actually destroying value: if Mohan's revenue is Rs1,500,000, accounting profit is a cheerful +Rs660,000, but economic profit is –Rs330,000 — he'd be *better off* back at his IT job renting out the building. Economic profit is the number that tells you whether to stay in business.

17. Family 13 — Cost Minimisation (Cobb–Douglas Production)

17.1 What the question asks

Given a production function and factor prices, find the cost-minimising input mix (and often the cost of a target output). Canonical (Jan 2026): $q = 450KL$, price of capital $r = \text{Rs}30/\text{day}$, price of labour $w = \text{Rs}150/\text{day}$.

17.2 Step-by-step

Minimise $C = rK + wL$ subject to producing q . The optimum equates the marginal product per rupee across inputs (the producer analogue of the consumer's bang-per-buck):

$$\frac{MP_K}{r} = \frac{MP_L}{w} \iff \text{MRTS} = \frac{MP_L}{MP_K} = \frac{w}{r}.$$

1. For $q = 450KL$: $MP_K = 450L$, $MP_L = 450K$.
2. $\text{MRTS} = \frac{MP_L}{MP_K} = \frac{450K}{450L} = \frac{K}{L}$.
3. Set $\text{MRTS} = \frac{w}{r}$: $\frac{K}{L} = \frac{150}{30} = 5 \Rightarrow \mathbf{K = 5L}$.

Labour is five times as expensive as capital (150 vs. 30), so the firm uses *five times as much capital as labour*. The cheaper input is used more intensively — exactly the substitution logic. To hit a specific output q , substitute $K = 5L$ into $q = 450KL = 450(5L)L = 2250L^2$, solve for L , then K , then cost $C = 30K + 150L$.

- Inverting the price ratio (using r/w instead of w/r).
- Confusing $\text{MRTS} = MP_L/MP_K$ with its reciprocal.
- Forgetting that this $q = 450KL$ form has *increasing* returns to scale (doubling both inputs quadruples output), unlike a constant-returns Cobb–Douglas.

18. Family 14 — Externalities (Beekeeper / Paper Mill)

18.1 What the question asks

A positive externality (beekeeper pollinates a neighbouring orchard) or a negative one (paper mill dumps effluent). Find the *private* quantity (ignoring the externality) and the *socially efficient* quantity (including it). Canonical (2025 T2 FN, beekeeper): $MC = 10 + 5Q$, each hive yields \$50 of honey, and each hive saves the orchard \$10 of artificial pollination.

18.2 Step-by-step (positive externality, beekeeper)

1. **Private optimum** (beekeeper ignores the orchard): produce while private marginal benefit $\geq$ marginal cost. Honey is worth \$50/hive, so $50 = 10 + 5Q \Rightarrow Q_{\text{private}} = 8$ hives.
2. **Social optimum** (add the \$10 external benefit per hive): social marginal benefit = $50 + 10 = 60$, so $60 = 10 + 5Q \Rightarrow Q_{\text{social}} = 10$ hives.

The beekeeper privately keeps 8 hives but the social optimum is 10 — a positive externality is **under-provided** by the market (the beekeeper isn't paid for the pollination he provides). A **Pigovian subsidy** of \$10 per hive (equal to the external benefit) would raise his private incentive to the social optimum. (Coase's famous point: the beekeeper and orchard owner could also *bargain* to the efficient outcome if transaction costs are low.)

18.3 Negative-externality variant (paper mill)

For the paper market ($Q_D = 160,000 - 2000P$, $Q_S = 40,000 + 2000P$, $MEC = 0.0006Q$):

1. **Competitive (private) equilibrium:** $160,000 - 2000P = 40,000 + 2000P \Rightarrow P = 30$, $Q = 100,000$.
2. **Social optimum:** demand price = private supply price + MEC. Inverse demand $P = 80 - \frac{Q}{2000}$; inverse supply $P = \frac{Q}{2000} - 20$; add $MEC = 0.0006Q$:

$$80 - \frac{Q}{2000} = \left(\frac{Q}{2000} - 20\right) + 0.0006Q \Rightarrow 100 = 0.0016Q \Rightarrow Q^* = 62,500.$$

3. The market over-produces ($100,000 > 62,500$); a **Pigovian tax** equal to the MEC at Q^* restores efficiency.

- Adding the externality to the wrong side (a *benefit* raises marginal benefit; a *cost* raises marginal cost).
- For positive externalities, expecting over-production — it's the opposite (under-production).
- Forgetting to convert $Q_D = a - bP$ into inverse form before equating to a marginal curve.

19. Family 15 — Signalling Decision (Get an MBA?) and Family 16 — Sunk Cost (Fritos)

19.1 Family 15 — the MBA signalling decision

What it asks (2025 T2 FN): an MBA costs 10 (including forgone income). Your future income depends on the labour market: strong $\Rightarrow$ MBA worth 32 vs. status-quo 12; average $\Rightarrow$ 16 vs. 8; weak $\Rightarrow$ (lower still). For which probability p of a strong market should you get the MBA?

Method. Get the MBA iff its *expected net* value beats the status quo:

$$\mathbb{E}[\text{value} \mid \text{MBA}] - 10 \geq \mathbb{E}[\text{value} \mid \text{status quo}].$$

Compute each expectation as a probability-weighted sum over market states, then solve the inequality for p . The MBA is worthwhile when p (probability of the strong market, where the MBA's premium is largest) is high enough to cover the cost 10.

The MBA's value here isn't only the skills — it's a **costly signal** (Week 11). It pays off most when the labour market rewards the signal (strong state). The threshold p is exactly where the *expected signalling premium* equals the *cost of signalling* (10). Below that p , the signal isn't worth its cost; above it, it is.

The exact payoff in the “weak” labour market (and the precise probability split) sit in the image layer of the paper. Set up the inequality with your paper's three payoffs and solve for the threshold p ; the *method* above is what's being tested. Verify the numbers with your instructor.

19.1.1 The follow-up: “if $p = 0.4$, the highest price the university can charge”

A second sub-question fixes the probability (e.g. $p = 0.4$) and asks for the *maximum tuition* the university could charge such that you still choose the MBA. Method: with p plugged in, compute the expected *benefit* of the MBA over the status quo, $\mathbb{E}[\text{MBA value}] - \mathbb{E}[\text{status-quo value}]$. You will pay for the MBA as long as the price does not exceed that expected benefit, so

$$\text{highest price} = \mathbb{E}[\text{MBA value}] - \mathbb{E}[\text{status-quo value}] \quad (\text{at the given } p).$$

That break-even price is exactly where the net gain from the signal falls to zero — charge a rupee more and you'd skip the MBA.

“For which p get the MBA?” solves the inequality for p (holding price = 10). “At $p = 0.4$, the highest price?” solves the *same* inequality for the *price* (holding p fixed). Both pin down the point where expected signalling benefit = cost.

19.2 Family 16 — the sunk-cost decision (Fritos)

What it asks (2024 T1): you paid \$100 for five cases of Fritos a customer rejected. Your grocer refunds \$0.25 per \$1 of returned goods (so \$25 salvage). A fraternity now offers \$55 for all five. Should you sell to the fraternity?

Solution. Yes — sell to the fraternity. Compare only the *forward-looking* options: \$55 (fraternity) vs. \$25 (return to grocer). $\$55 > \25 , so sell to the fraternity. The \$100 already paid is a **sunk cost** — irreversible and irrelevant to the decision.

The opportunity-cost sub-question. The paper also asks: “what is the opportunity cost of filling this order (selling to the fraternity)?” The answer is **\$25** — the salvage value you *give up* by not returning the cases to the grocer. Opportunity cost is the value of the *best forgone alternative*, which here is the \$25 refund, *not* the \$100 you paid. (You’d accept the fraternity’s offer because \$55 exceeds this \$25 opportunity cost.)

A sunk cost cannot be recovered and must be *ignored* in any decision. The only question is which *future* option yields more. The \$100 feels like it should matter (“I can’t sell \$100 of Fritos for \$55!”), but it’s gone either way — selling for \$55 beats salvaging for \$25 by \$30. Letting sunk costs drive choices is the **sunk-cost fallacy**.

Family 15 (MBA): “for which $p \dots$ ” $\Rightarrow$ set expected-value inequality, solve for the threshold probability. Family 16 (Fritos): the instant you see a *past, unrecoverable* payment, label it sunk and *delete it* from the comparison — then pick the larger future payoff.

20. Family 17 — Consolidated Conceptual MCQ Bank

20.1 What the question asks

Each paper carries a block of one-line conceptual MCQs. Here is the consolidated bank with answers and justifications, grouped by topic. These are guaranteed marks if memorised.

20.1.1 Market structure

Statement	Answer & why
“True about monopolistic competition”	Many differentiated firms; free entry; long-run $P = ATC$ (zero economic profit); $P > MC$ (markup); excess capacity.
“A monopolist always produces in...”	the elastic region of demand ($ \varepsilon > 1$), where $MR > 0$.
“Cobb–Douglas $F(K, L) = K^p L^q \dots$ ”	May show increasing, decreasing, or constant returns to scale (depending on whether $p + q \gtrless 1$). Not constant MP.

20.1.2 Public goods, common resources, externalities

Statement	Answer & why
“A common resource is...”	rival and non-excludable (e.g. ocean fish): your use subtracts from others’, but no one can be excluded.
“Public goods are provided by the government because...”	private firms can’t profit (free-rider problem makes them non-excludable, so markets under-provide them).
“Which goods represent network goods?”	Twitter, Microsoft Excel, Facebook — their value rises with the number of other users (network effects).

20.1.3 Information

Statement	Answer & why
“Adverse selection occurs when...”	a hidden <i>type</i> (a risky person buys insurance) distorts the market <i>before</i> the deal.
“Moral hazard occurs when contracts are written so that...”	an insured/covered party takes a hidden <i>action</i> (more risk, less care) <i>after</i> the deal.
“If bad drivers avoid tickets, insurers will...”	lose the driving record as a screening device (it no longer separates types).

20.1.4 Consumer theory revision (carried into the End-Term)

Statement	Answer & why
“Income rises and one price falls at the same time $\Rightarrow$ the consumer is <i>definitely</i> better off.”	True — the budget set strictly expands, so the optimum is at least as good (weakly dominates), in fact strictly better here.
“A firm can always trade 3 units of labour for 1 of capital and keep output constant.”	Perfect substitutes in production (constant MRTS = 3) — linear isoquants.
“Tariff results in. . .”	higher domestic price, lower consumption, higher domestic production, government revenue, and a deadweight loss (consumption + production distortions).

Anchor each to a definition: *rival/excludable* grid for goods; *hidden type vs. hidden action* for selection vs. hazard; *informed vs. uninformed mover* for signalling vs. screening; *network effect* = value rises with users; *returns to scale* depends on $p + q$ in Cobb–Douglas. If you can state the definition, the MCQ answers itself.

Appendices

21. Appendix A — Master Formula Sheet (Weeks 9–12)

- Cournot reaction: $q_i = \frac{a-c}{2b} - \frac{q_j}{2}$; equilibrium $q_i = \frac{a-c}{3b}$, total $Q = \frac{2(a-c)}{3b}$.
- n -firm Cournot: $q = \frac{a-c}{b(n+1)}$, $Q = \frac{n}{n+1} \frac{a-c}{b}$.
- Bertrand (homogeneous): $P = MC$ with two firms.
- Stackelberg: leader $\frac{a-c}{2b}$, follower $\frac{a-c}{4b}$, $Q = \frac{3(a-c)}{4b}$.
- Monopoly: $Q = \frac{a-c}{2b}$. Ranking Q : monopoly < Cournot < Stackelberg < PC.
- Collusion $\rightarrow$ monopoly outcome (unstable, illegal). Hotelling $\rightarrow$ median location.

- $EV = \sum p_i x_i$; $Var = \sum p_i (x_i - EV)^2$; $\sigma = \sqrt{Var}$; $CV = \sigma / EV$.
- Decision rules under risk: max EV; mean-variance; min coefficient of variation.
- Expected utility $EU = \sum_i \pi_i u(c_i)$ (additively separable across states).
- Risk averse $u'' < 0$ ($u(EV) > EU$); neutral linear; loving $u'' > 0$.
- Certainty equivalent: $u(CE) = EU$. Risk premium = $EV - CE$.
- Fair insurance premium $M = B(1-p)$; risk-averse $\Rightarrow$ full insurance.
- Advertising thumb-rule: gross gain $P \Delta Q = \varepsilon$.
- Uncertainty rules: maximax, maximin, minimax regret, equal-probability.

- Adverse selection (hidden type): peaches trade only if $6 - 4p \geq 5$, i.e. $p \leq \frac{1}{4}$ (lemons example); market can unravel.
- Moral hazard (hidden action): principal-agent; mitigate via incentive contracts (fixed wage / licence / bonus / option / commission — trade off incentives vs. risk), monitoring, bonding, deferred pay, efficiency wages, co-pays.
- Signalling (informed party, costly signal) vs. screening (uninformed party, device).
- Externality: $SMC = PMC + EMC$ (neg., over-produce); $SMB = PMB + EMB$ (pos., under-produce). Pigovian tax = EMC ; subsidy = EMB ; or tradable permits.
- Public good = non-rival + non-excludable; **Samuelson rule** $\sum_i MB_i = MC$ (vertical sum); free-rider problem.
- Network effects $\rightarrow$ monopoly tendency, chicken-and-egg, lock-in. Platforms: same-side (direct) vs. cross-side (indirect) externalities.

22. Appendix B — Glossary of Key Terms

- **Cournot / Bertrand / Stackelberg:** quantity-simultaneous / price / quantity- sequential oligopoly models.
- **Reaction (best-response) function:** a firm's optimal output given rivals'.
- **Backward induction:** solving a sequential game from the end.
- **Credible threat:** a threat the threatener would actually carry out.
- **Braess's paradox:** adding a route can worsen everyone's travel time.
- **Monopolistic competition:** many differentiated firms, free entry, LR $P = ATC$.
- **Expected utility:** probability-weighted utility across states.
- **Certainty equivalent / risk premium:** sure amount as good as a gamble / the EV given up to avoid risk.
- **Actuarially fair insurance:** premium = probability $\times$ benefit.
- **Adverse selection:** market distortion from hidden *type*.
- **Moral hazard:** distortion from hidden *action*.
- **Signalling / screening:** costly action by the informed / device by the uninformed party.
- **Externality:** uncompensated effect on a bystander.
- **Pigovian tax:** tax equal to the external marginal cost.
- **Tradable permit:** a tradeable right to pollute up to a cap (carbon trading).
- **Public good:** non-rival, non-excludable. **Free rider:** enjoys without paying.
- **Samuelson rule:** provide a public good until $\sum MB_i = MC$.
- **Network effect:** value rising with the number of users.
- **Platform / two-sided market:** intermediary connecting two user groups.
- **Lock-in:** being stuck with an inferior standard due to network effects.

23. Appendix C — One-Page Exam-Day Checklist

Before you start solving

1. **Oligopoly?** Identify the model. Cournot $\Rightarrow$ reaction functions & $\frac{a-c}{3b}$. Stackelberg $\Rightarrow$ backward induction, leader $\frac{a-c}{2b}$. Bertrand $\Rightarrow P = MC$.
2. **Monopoly?** $MR = MC$ (twice-slope MR); Lerner = $\frac{P-MC}{P} = \frac{1}{|\epsilon|}$.
3. **Risk?** Run incomes through u first. Compare $u(\text{sure})$ vs. EU . CE from $u(CE) = EU$; risk premium = $EV - CE$.
4. **Insurance?** Fair premium $M = B(1 - p)$; risk-averse $\Rightarrow$ full cover. Adverse-selection market: check who participates at the premium.
5. **Information question?** Timing test: hidden type before = adverse selection; hidden action after = moral hazard. Informed acts = signalling; uninformed acts = screening.
6. **Externality?** Draw private vs. social curves; over/under-production; Pigovian tax/-subsidy = external marginal cost/benefit.

7. **Public good?** Vertical summation (Samuelson); watch the free-rider logic.
8. **Platform?** Network effects, same-side vs. cross-side, subsidise one side.

This completes the three-volume Managerial Economics e-book. Volume III pulls together the strategic and informational frontier of the course — oligopoly, risk, asymmetric information, externalities, public goods, and platforms — the tools a data scientist needs to reason about real markets. Re-derive the boxed formulae from scratch, rehearse the PYQ recipes until automatic, and you'll walk into the End-Term with confidence. All the best.

Made by Anmol Kansal with AI — Managerial Economics, Volume III (Weeks 9–12).